

Labor Market Reforms in Open Economies: Current Account Dynamics and Consumer Heterogeneity[†]

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January 2026

Abstract

This paper links labor market reforms to external balances through precautionary saving under imperfect insurance against unemployment risk. We quantify the mechanism for a major German benefit reform by combining fixed-effects evidence from the German Socioeconomic Panel with a small open economy heterogeneous agent model featuring search frictions. The reform-induced rise in private saving permanently increases net foreign assets. The model accounts for up to 12 percent of post-reform current account dynamics and delivers empirically consistent real exchange rate movements, whereas these dynamics are largely absent under complete insurance. Welfare effects are highly heterogeneous across employment status, skill, and wealth.

JEL classification: E21, E24, F16, F41.

Keywords: Heterogeneous Agents, Unemployment Benefit Reform, Precautionary Savings.

[†] We thank Adrien Auclert, Christian Bayer, Corina Boar, Benjamin Born, Jeanne Commault, Sergio de Ferra, Jeppe Druedahl, Christian Haefke, Philip Jung, Keith Kuester, Ralph Lütticke, Ioana Marinescu, Jochen Mankart, Christian Merkl, Kurt Mitman, Morten Graugaard Olsen, Paul Pichler, Timo Sauerbier, Ayşegül Şahin, Ctirad Slavik, Søren Hove Ravn, Morten Ravn, John Kramer, and Rune Vejlin for their valuable comments. We also received valuable feedback during seminars and presentations at the University of Kiel, University of Konstanz, the Macroeconomic Policies with Heterogeneous Agents Workshop (2024) in Paris, T2m (2022), and the Austrian Economic Association (2022). Brigitte Hochmuth gratefully acknowledges funding from the German Research Foundation (DFG) under grant number ME-3887/5-1. She also thanks the University of Bonn for its hospitality in 2023/2024 while writing this paper, during which she benefited from its research infrastructure. The views expressed in this paper are those of the authors and do not necessarily reflect the views of the Deutsche Bundesbank or its staff.

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1 Introduction

How do economies respond to structural labor market reforms, and what role do precautionary savings play when insurance against unemployment risk is incomplete? While past studies have explored these dynamics in closed economies using heterogeneous agent models, a significant gap exists in understanding their impact in open economies, particularly regarding the effects on external balances of reducing unemployment benefits.¹ In contrast, studies that evaluate labor market reforms in open economies typically rely on representative agent models, which miss the incentive to accumulate precautionary savings due to perfect consumption insurance.²

This paper fills this gap by introducing a small open economy model with incomplete insurance and a frictional labor market. Our model includes a detailed unemployment benefit system and realistic wealth inequality to capture the international effects of labor market reforms.³ Our analysis links the reduction in unemployment benefit generosity to the build-up of net foreign assets (NFA) through a *precautionary savings channel*. As a case study, we examine the German Hartz IV reform, which cut long-term unemployment benefits in 2005 and shortened short-term benefits in 2006.⁴

Germany's experience is a useful case study for understanding how domestic policy changes can interact with international macroeconomic outcomes. From the early 2000s until 2019, Germany recorded a persistently positive current account balance (see Figure 1), alongside a sizable rise in its net foreign asset position and a sustained gap between aggregate saving and investment (see Figure 13 in Appendix A.1).⁵ These external outcomes have been the focus of substantial debate about the underlying mechanisms and policies. One frequently cited hypothesis links the Hartz IV reforms to the evolution of the current account and net foreign assets via labor market adjustments and savings channels.⁶ While macroeconomic studies have quantified important *domestic* effects of these reforms,⁷ their *international* consequences and the general equilibrium channels linking saving, investment, and the current account have received less attention.

We proceed in two steps. First, using household fixed-effects regressions with data from the German Socioeconomic Panel (GSOEP), we estimate that the reform raised the aggregate savings rate by about one percentage point. Savings increased more strongly for workers more exposed to the entitlement cut, while civil servants, who face almost no unemployment risk in Germany, show no corresponding change in savings behavior. Second, we discipline our quan-

¹ The role of precautionary savings due to incomplete insurance against the risk of becoming unemployed has been stressed by [Krusell, Mukoyama, and Şahin \(2010\)](#), [Krueger, Mitman, and Perri \(2016b\)](#), [Krause and Uhlig \(2012\)](#), [Krebs and Scheffel \(2013\)](#), [Launov and Wälde \(2013\)](#), and [Nakajima \(2012\)](#) among others.

² See [Cacciatore, Duval, Fiori, and Ghironi \(2016\)](#).

³ An earlier version of this paper circulated under the title "Labor Market Reforms, Precautionary Savings, and Global Imbalances". The current version fundamentally differs from the earlier version, featuring full consumer heterogeneity in a small open economy.

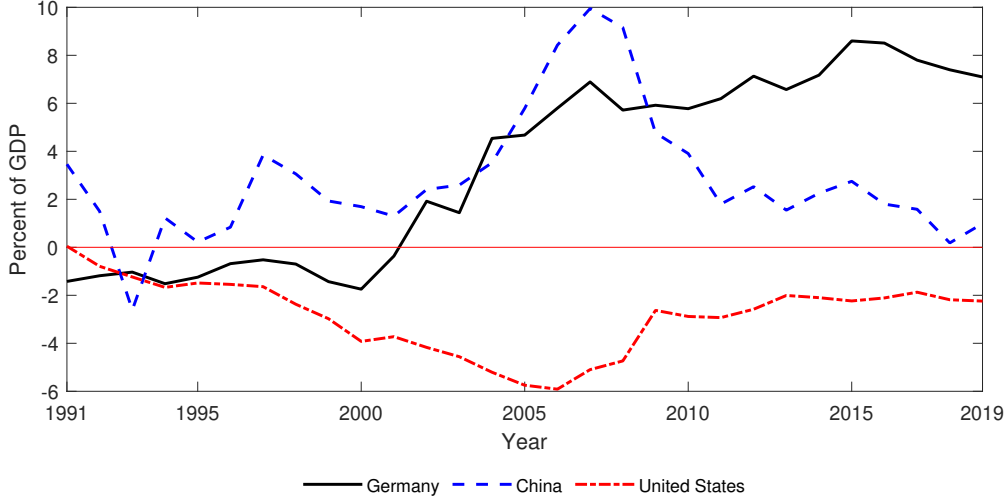
⁴ A detailed description of the Hartz reforms and the developments of the German current account, as well as its net foreign asset position, can be found in Appendix A.

⁵ See, e.g., [The Economist \(2017b\)](#) and [International Monetary Fund \(2020\)](#). For a discussion of capital exports and returns, see [Hünnekes, Konradt, Schularick, Trebesch, and Wingenbach \(2025\)](#).

⁶ See, e.g., [Krugman \(2014\)](#), who argues that wage restraint and cost competitiveness after the reforms contributed to Germany's trade surpluses.

⁷ See [Krause and Uhlig \(2012\)](#), [Krebs and Scheffel \(2013\)](#), [Launov and Wälde \(2013\)](#), [Hochmuth, Kohlbrecher, Merkl, and Gartner \(2021\)](#), and [Hartung, Jung, and Kuhn \(2025\)](#).

Figure 1: Current Account in percent of GDP for Germany, China, and the United States, 1991-2019



Notes: The figure depicts the Current Account in percent of GDP for Germany (black, solid line), China (blue, dashed line), and the United States (red, dashed line) for the years 1991 to 2019. Data source: The World Bank Data (2021).

titative model by matching this empirical savings rate response: we generate simulated panel data mirroring the empirical sample and re-estimate the same fixed-effects regressions on the simulated data, targeting the one-percentage-point post-reform increase in aggregate saving. As a partial-equilibrium benchmark, a back-of-the-envelope mapping of the micro-based savings response implies additional net saving of roughly 0.4 percent of GDP per year, corresponding to an average contribution of about 6 percent of the observed current account balance. The most affected group accounts for only a limited share of this direct saving channel. The implied magnitude largely stems from modest yet broad-based changes in saving rates among working-age households in the GSOEP.

In general equilibrium, the calibrated model implies that the reform materially affected Germany's external adjustment, accounting for up to 12 percent of the observed change in the current account over the first five years, alongside non-negligible movements in the real exchange rate and unemployment. We benchmark these outcomes against complete-insurance and closed-economy versions of the model, highlighting where the complete-insurance framework departs from the dynamics above.

Our model of a small open economy accounts for consumer heterogeneity in asset holdings, employment status, and skills. The labor market is subject to matching frictions (Pissarides, 2000) and features a two-tier unemployment insurance system with distinct short-term and long-term tiers, mirroring German institutions and the reform's differential changes across unemployment durations. Workers enter the labor force as low-skilled, accumulate skills over time, and have wages negotiated by a labor union for both low- and high-skilled workers. Risk-averse households smooth consumption over their lifetime. A cut in unemployment benefits raises consumption risk upon job loss when insurance is incomplete. Households increase precautionary savings, substituting away from public insurance toward self-insurance. In line with the empirical focus, the quantitative analysis focuses on saving *flows* of working-age households and their im-

plications for external adjustment.⁸ In the open economy, part of this higher desired saving is absorbed through foreign bond holdings, implying a higher net foreign asset position in equilibrium.

We emphasize the importance of consumer heterogeneity in evaluating the macroeconomic consequences of labor market reforms by comparing model responses to those in a complete insurance framework. Our model features a precautionary savings channel that is absent in canonical small open economy representative agent models (see [Obstfeld and Rogoff, 1995](#)). These models typically highlight two opposing mechanisms in response to reduced unemployment benefits: a relative-price/expenditure-switching channel that can raise net exports during the transition, and a demand channel in which higher employment raises aggregate consumption. We show that, in the absence of a precautionary savings motive, the demand response dominates, and the reforming country's current account declines. Introducing incomplete insurance fundamentally changes the model dynamics and yields *qualitatively different* current account responses, highlighting the broader relevance of self-insurance in evaluating labor market reforms beyond the German case. In our framework, a reduction in the generosity of unemployment benefits raises consumption risk upon job loss and induces households to self-insure by increasing desired savings. In the small open economy, part of this additional saving is absorbed through foreign bond holdings, generating temporary current account surpluses and a rising net foreign asset position. At the same time, the wage decline triggers a real exchange rate depreciation and expenditure switching toward domestic goods, which raises net exports early in the transition. These forces are dampened by improved labor market conditions, which reduce unemployment risk and the incentive to save, and by higher investment and domestic demand, which raise imports over time. Quantitatively, the net saving response dominates. External assets rise, leaving the economy with a permanently higher *NFA* position in the new steady state.⁹ We show that the incomplete-insurance model matches the post-reform dynamics of key macroeconomic aggregates more closely and reproduces the empirically observed directions of adjustment in the current account and the net foreign asset position. Consumer heterogeneity is also central for the distributional implications of the reform. In particular, low-skilled households raise precautionary saving and reduce their reliance on borrowing constraints, which increases asset holdings for initially wealth-poor groups in the post-reform steady state. As a result, the model implies a decline in overall wealth inequality.

Welfare effects of the reform are highly heterogeneous, with the most significant losses concentrated among low-skilled and unemployed workers, particularly those with low asset holdings. These agents rely heavily on labor income and have a high marginal propensity to consume, making them especially vulnerable to the reduction in unemployment benefits. To self-insure, they are forced to reduce current consumption, leading to a substantial decline in their welfare. In contrast, high-skilled and asset-rich individuals can more easily smooth consumption and mitigate the negative effects of the reform, with some even experiencing modest long-run welfare gains. Under complete insurance, the reform leads to uniform welfare gains, as improved la-

⁸ Accordingly, the framework is not intended as a model of wealth concentration at the very top of the distribution.

⁹ As net foreign income increases at this higher *NFA* level and the current account returns to zero in steady state, net exports adjust downward.

bor market conditions allow agents to benefit from higher job-finding rates without experiencing consumption losses. Additionally, comparing the small open economy with a closed-economy setting highlights an additional difference: while the small open economy experiences aggregate welfare losses due to stronger precautionary savings behavior and a slower consumption recovery, the closed economy benefits from a temporary rise in the interest rate and stronger labor market responses, yielding positive average welfare effects.

Related Literature: This paper is connected to multiple areas of the literature. First, it relates to studies discussing the link between labor market reforms and current account disparities. [Kollmann, Ratto, Roeger, in 't Veld, and Vogel \(2015\)](#) find that the Hartz reforms were one of the main drivers of Germany's external balance. They abstract from modeling a frictional labor market and instead interpret shocks to leisure as changes in the generosity of unemployment benefits. This is confirmed by [Ruppert and Stähler \(2020\)](#) in a three-region life-cycle model with labor market frictions, but contrasts with [Gadatsch, Stähler, and Weigert \(2016\)](#). [Cacciatore et al. \(2016\)](#) study the effects of labor market deregulations in general and find quantitatively small effects on the current account. These studies feature perfect insurance; thus, there is no precautionary savings motive. We contribute to this strand of the literature by allowing for consumer heterogeneity and a precautionary savings motive.¹⁰

Empirical studies on the impact of labor market reforms on the current account include [Kennedy and Slok \(2005\)](#) and [Bertola and Lo Prete \(2015\)](#). [Kennedy and Slok \(2005\)](#) show that labor market deregulation initially reduces prices and wages, raising the trade balance, but eventually reverses the current account due to increased foreign capital inflows driven by higher domestic profitability. [Bertola and Lo Prete \(2015\)](#) find that labor market deregulations generally increase a country's current account, emphasizing the role of precautionary savings, though their focus is on financial market imperfections and human capital investment. Our work complements [Bertola and Lo Prete \(2015\)](#) by quantifying the effects of specific deregulations and highlighting the role of idiosyncratic consumption risk.

Second, our work relates to the literature that addresses uninsurable labor market risk with precautionary savings, such as [Challe and Ragot, 2016](#), [Challe, Matheron, Ragot, and Rubio-Ramirez, 2017](#), [Krusell et al., 2010](#), [Heathcote and Perri, 2018](#), as well as [McKay and Reis, 2016](#), [McKay, 2017](#), [McKay, Nakamura, and Steinsson, 2017](#), and [Ravn and Sterk, 2017](#).¹¹ [Fernandes and Rigato \(2024\)](#) empirically assess the precautionary saving channel using variations in unemployment insurance across U.S. states and over time, finding small and statistically non-significant effects consistent with a present-biased model. The effects in our paper differ from those in [Fernandes and Rigato \(2024\)](#) because the Hartz IV reform was a permanent structural change, whereas their analysis examines temporary, countercyclical unemployment insurance extensions, and because Germany's higher pre-reform replacement rates made the benefit re-

¹⁰ [Hochmuth, Merkl, and Stüber \(2024\)](#) extend this discussion by examining how reduced unemployment benefits in one country can generate significant long-run consumption spillovers to non-reforming countries under incomplete insurance.

¹¹ Further related are [Krueger, Mitman, and Perri \(2016a\)](#) and [Krueger et al. \(2016b\)](#) who focus on the role of unemployment insurance systems in analyzing welfare and consumption effects in heterogeneous agent models. [Bayer, Luetticke, Pham-Dao, and Tjaden \(2019\)](#) study the effects of income uncertainty on households' precautionary savings.

duction more pronounced.¹² These papers are based on a closed-economy framework and, therefore, are silent on the effects in an international dimension.

Third, we contribute to the recent emerging literature analyzing the effects of (mostly foreign) shocks in heterogeneous-agent open-economy models. [de Ferra, Mitman, and Romei \(2020\)](#) highlight the importance of household portfolio composition in the transmission of foreign shocks, while [Auclert, Rognlie, Souchier, and Straub \(2021\)](#) and [Oskolkov \(2021\)](#) explore how heterogeneous households respond to exchange rate depreciations and external monetary shocks, respectively. [Guo, Ottonello, and Perez \(2020\)](#) and [Zhou \(2022\)](#) further investigate the role of financial integration and redistribution in shaping the distributional impacts of these shocks. Additionally, [Aggarwal, Auclert, Rognlie, and Straub \(2023\)](#) analyze the effects of deficit-financed fiscal transfers, showing how they lead to excess savings and persistent current account deficits.¹³ While these studies focus on the distributional effects of both foreign and domestic shocks, our work specifically examines the current account implications of a permanent domestic policy reform in the absence of nominal rigidities.

Fourth, we contribute to the literature on current account disparities in quantitative macroeconomic models. [Mendoza, Quadrini, and Rios-Rull \(2009\)](#) study the role of financial market developments and financial integration across countries in explaining global imbalances. [Caballero, Farhi, and Gourinchas \(2008\)](#) explain these disparities through heterogeneity in a country's ability to generate financial assets for global savers, and [Hunt and Rebucci \(2005\)](#) analyze the role of accelerating productivity growth. [Broer \(2014\)](#) highlights how rising income risk can endogenously relax borrowing constraints, leading to reductions in aggregate foreign assets even in the presence of precautionary savings motives. [de Ferra, Mitman, and Romei \(2021\)](#) identify a negative relationship between a country's income inequality and its current account balance, which they rationalize in a model with endogenously incomplete financial markets. Further related are [Hoffmann, Krause, and Tillmann \(2019\)](#), who show that higher uncertainty in income streams increases precautionary savings and leads to a higher long-run external asset position. [Carroll and Jeanne \(2009\)](#) introduce a tractable framework of net foreign assets, where the level of domestic wealth is determined by precautionary savings in the presence of idiosyncratic shocks. Finally, our paper is related to studies evaluating the effects of the Hartz IV reform on German unemployment from a macroeconomic perspective (see e.g. [Krebs and Scheffel, 2013](#), [Krause and Uhlig, 2012](#), [Launov and Wälde, 2013](#), [Hochmuth et al., 2021](#), [Hartung et al., 2025](#)). These papers all evaluate the reform effects in a closed economy setting.

The rest of the paper is structured as follows. The next section presents empirical evidence on how the reform affected German savings rates. Section 3 derives an open-economy heterogeneous agent model with search and matching frictions. We explain the calibration in Section 4. Section 5 shows our results. We discuss the effects of consumer heterogeneity in Section 6 and analyze welfare effects in Section 7. Section 8 concludes.

¹² The initial replacement rate before Hartz IV was significantly higher in Germany — approximately 67 percent of previous net earnings, compared to 40 percent in the U.S.

¹³ Recent work by [Bayer, Kriwoluzky, Müller, and Seyrich \(2024\)](#) shows that in a Heterogeneous Agent New Keynesian (HANK) model of a two-country monetary union, business cycle shocks primarily redistribute across countries rather than within the union-wide wealth distribution.

2 Empirical Evidence

In this section, we present descriptive evidence and household-level fixed effects regression results on the impact of the Hartz IV labor market reform on German households' savings behavior, using data from the German Socioeconomic Panel (GSOEP).

Institutional Background: In response to rising unemployment and low economic growth rates in the late 1990s and early 2000s (when Germany was referred to as the "sick man of Europe", see [The Economist, 2017a](#)), the Hartz reforms were implemented gradually from 2003 to 2006. These reforms aimed to reduce unemployment, enhance labor-market efficiency, and increase the incentives to work by reducing the generosity of unemployment benefits. Appendix A.2 provides further institutional details on the Hartz reforms. The Hartz IV reform, introduced in 2005, marked a major overhaul of Germany's social security system. Before the reform, unemployment support consisted of three tiers: (i) short-term unemployment benefits (up to 67% of previous net earnings for a maximum of 32 months), (ii) long-term benefits (up to 57% of previous net earnings, potentially until retirement), and (iii) social assistance for those ineligible for unemployment benefits. Hartz IV consolidated long-term unemployment benefits and social assistance into a single program, "Unemployment Benefits II" (Arbeitslosengeld II), which provided a flat-rate benefit independent of prior earnings, set at 345 Euros for a single household, with possible top-ups for housing, heating, and additional living expenses. In a second reform step in 2006, the entitlement period for short-term unemployment benefits was reduced, with the extent of the reduction dependent on age. Long-term unemployment benefits in Germany are subject to means testing. However, the asset limits are generally not restrictive enough to significantly distort the savings decisions of most households. Data from the Deutsche Bundesbank and the Institute for Employment Research (IAB) show that only a small percentage of households exceed these thresholds (see Appendix A.2), indicating that the majority are not adversely impacted when qualifying for benefits. A detailed discussion of the role of asset limits in influencing savings behavior is provided in Appendix A.2.

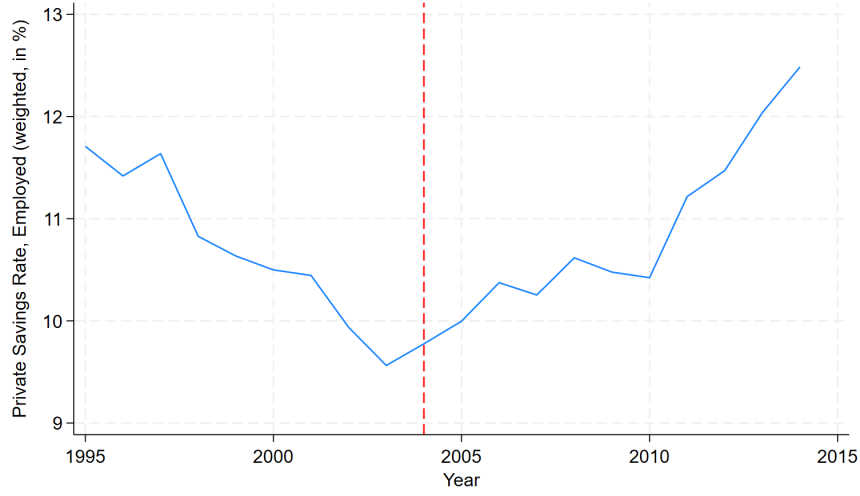
Descriptives: As private households constitute the largest component of the German current account (see Figure 14 in Appendix A), shifts in their savings behavior can have significant implications for the overall current account balance. Figure 2 depicts the savings rate for employed household heads based on microeconomic data from the German Socioeconomic Panel (GSOEP).¹⁴ Before the reform, the private savings rate showed a declining trend. By 2005, when the unemployment benefit reform was implemented, we observe a reversal of this trend and an increase in the private savings rate.¹⁵

Fixed-Effects Regression: In our subsequent analysis, we examine how the aggregate savings rate and the savings rates of the most affected groups changed in response to the reform. To

¹⁴ The GSOEP data does not provide information on the amount of precautionary savings (e.g. due to labor market risk) for this period. For details on the GSOEP and the savings rate calculation, see Appendix B.

¹⁵ Figure 15 in Appendix B.1.1 illustrates a trend reversal in savings rates across different educational groups, with low-educated workers initially increasing their savings rates strongly post-reform.

Figure 2: Private Savings Rate of Employed Workers, 1995-2014



Notes: We calculate the private savings rate for employed workers based on household-level survey data of the German Socioeconomic Panel (GSOEP). The calculation follows [Stein \(2009\)](#); for details, see Appendix B. The dashed vertical line denotes the year 2004 (right before the implementation of the reform).

identify the most affected groups, we build on the classification provided by [Hartung et al. \(2025\)](#), which distinguishes variations in the reduction of maximum benefit duration based on age and prior employment duration. The largest decrease in entitlement duration was observed among workers aged 45 to 55 with over 3 years of prior employment. Furthermore, we refine our analysis by dividing affected workers into subgroups: "Group 1" encompasses those who faced an entitlement cut of up to 20%, "Group 2" experienced a cut ranging from 21% to 40%, and "Group 3" underwent a reduction exceeding 41%.¹⁶

To assess the effect on the aggregate savings rate, we estimate the following household-level fixed effects model:

For the aggregate effect of the reform on the savings rate:

$$SavingsRate_{it} = \beta_0 + \beta_1 HartzIVDummy_t + \mathbf{X}_{it}\gamma + \mathbf{Z}_t\delta + \mu_i + \epsilon_{it}, \quad (2.1)$$

where $SavingsRate_{it}$ is the savings rate of household i at time t , and $HartzIVDummy_t$ equals 1 for the post-reform period (2005 onward). $\mathbf{X}_{it}$ includes household-level controls, and $\mathbf{Z}_t$ contains macroeconomic controls (i.e., a linear time trend, unemployment rate, GDP growth, interest rate, debt-to-GDP ratio, terms of trade, primary balance, consumer confidence, and the real effective exchange rate). μ_i denotes household fixed effects.

To capture the incremental effects on the most affected groups, we use:

$$SavingsRate_{it} = \beta_0 + \beta_1 MostAffected_{it} + \beta_2 (MostAffected_{it} \times HartzIVDummy_t) + \mathbf{X}_{it}\gamma + \lambda_t + \mu_i + \epsilon_{it}, \quad (2.2)$$

where $MostAffected_{it}$ is a time-varying indicator for households most affected by the Hartz reforms. The interaction term $(MostAffected_{it} \times HartzIVDummy_t)$ measures differential effects on

¹⁶ Groups 1–3 are not strict subgroups of the most affected workers but rather cover a broader range of individuals who experienced varying degrees of entitlement reductions.

these households. Here, time-fixed effects (λ_t) replace macroeconomic controls. We also estimate (2.2) for different groups based on the severity of entitlement cuts.

Table 1 presents the results of household-level fixed-effects estimations. Column (1) shows a statistically significant one-percentage-point increase in the aggregate savings rate post-reform, suggesting that a substantial portion of the private savings rate increase in Figure 2 can be attributed to the reform. Columns (2) and (3) refine the analysis by estimating the incremental effects on the most affected groups. These specifications replace macroeconomic control variables with time-fixed effects to account for common year-specific shocks affecting all households.¹⁷ In column (2), the interaction between "Most Affected" and the "Hartz IV-Dummy" indicates a significant rise in savings rates among the most impacted groups. Column (3) shows that the post-reform incremental increase in savings, captured by the group-specific Hartz IV interaction terms, is larger for larger entitlement cuts (especially for Groups 2 and 3).¹⁸ In the regressions, we control for potential time-varying household-level characteristics, i.e., marital status, region (East/West Germany), the number of children in the household, employment level, and ISCO-08 occupation.

Figure 3 shows the interaction terms of year dummies with the most affected workers, along with 90% confidence bands, capturing the dynamic effects of changes in the savings behavior. Before the reform, the year-dummy interactions for the most affected group are small and close to zero relative to the reference group. From 2005 onward, the estimates turn positive and significant and remain elevated in subsequent years. This suggests that the Hartz IV reform significantly changed the savings behavior of the most affected workers in the years after the reform. In contrast, there is no evidence that the 2001 pension reform, including the introduction of the "Riester-Rente" (a state-subsidized third pillar of the pension system), significantly affected the savings behavior of these workers in the early 2000s. While the Riester system was introduced to encourage private retirement savings, its uptake remained limited, particularly among low-income individuals (see [Geyer, Grabka, and Haan, 2021](#)).

Robustness: Civil Servants To validate our micro-level estimates of the incremental effects on savings rates, we examine the savings behavior of civil servants with characteristics similar to those of the most affected workers (along the dimensions of age and employment spell). In Germany, civil servants face minimal unemployment risk, as their employment contracts can be terminated only in exceptional circumstances.¹⁹ Consequently, we would not expect any significant shift in their savings behavior following the labor market reform. This is confirmed in Figure 4, which shows no notable change in their savings rates after 2005.

Back of the Envelope Calculation: Table 2 reports two exercises that quantify how much the Hartz IV reform can account for the rise in household saving and the associated change in Ger-

¹⁷ As a result, the Hartz IV-shift dummy drops out.

¹⁸ The reference category is Group 0, which includes workers younger than 45 with employment durations under 28 months, who experienced no change in entitlement duration.

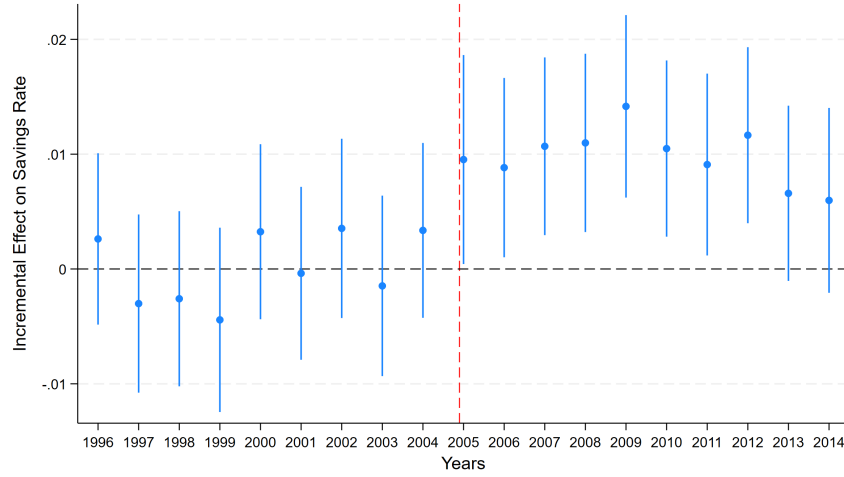
¹⁹ In Germany, civil servants can be dismissed for reasons such as serious misconduct, health issues, the abolition of their position, incompetence, violations of loyalty obligations, financial misconduct, and persistent non-performance, although the process is complex and highly regulated.

Table 1: Reform Effect on Saving Rates: Aggregate and Most Affected Groups

	(1)	(2)	(3)
<i>Dependent Variable: Saving rate</i>			
Hartz IV-Dummy	0.010*** (0.002)		
Most affected		-0.011*** (0.002)	
Most affected # Hartz IV-Dummy		0.009*** (0.002)	
Group 1			0.007** (0.003)
Group 2			-0.005*** (0.002)
Group 3			-0.007*** (0.002)
Group 1 # Hartz IV - Dummy			0.003 (0.002)
Group 2 # Hartz IV - Dummy			0.007*** (0.002)
Group 3 # Hartz IV - Dummy			0.011*** (0.002)
HH-level Controls	Yes	Yes	Yes
Macro Controls	Yes	No	No
HH Fixed Effects	Yes	Yes	Yes
Time Fixed Effects	No	Yes	Yes
R-squared	0.574	0.575	0.575
Observations	127898	127898	127782

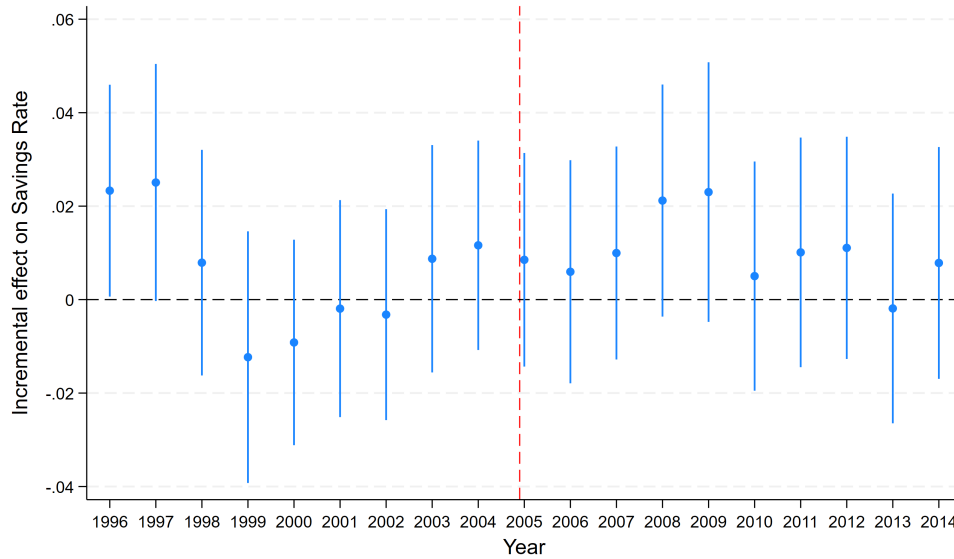
Note: "Most affected" denotes workers with an employment spell of min. 36 months and between the ages of 45 and 55. "Group 1" denotes workers who experienced an entitlement cut of up to 20%, "Group 2" denotes workers with an entitlement cut between 21% and 40%, and "Group 3" consists of workers with an entitlement cut of more than 41%. The entitlement cut varies by employment duration and age, see [Hartung et al. \(2025\)](#). "Hartz IV-Dummy" is a shift dummy that takes the value 1 from 2005 onward. The annual sample ranges from 1995 to 2014. All Regressions include the following household-level controls: Marriage Status, Region (East/West Germany), ISCO-08 occupation, number of children, and employment level. Specification (1) includes the following macroeconomic controls: a linear time trend, the unemployment rate, GDP growth, the interest rate, the debt-to-GDP ratio, the terms of trade, the primary balance, consumer confidence, and the real effective exchange rate (based on unit labor costs). Data Sources: GSOEP, Destatis, OECD, and Deutsche Bundesbank. Standard Errors are clustered at the household level. *p<0.1; **p<0.05; ***p<0.01

Figure 3: Incremental Effect on Savings Rate of Most Affected Group



Notes: We illustrate the point estimates of the interaction terms of year-dummies with the group of most affected workers together with 90% confidence intervals based on the household-level fixed effects regression that includes the following household-level controls: Marriage Status, Region (East/West Germany), ISCO-08 occupation, number of children, and employment level. The red vertical line illustrates the time of the reform implementation.
Data Source: German Socioeconomic Panel.

Figure 4: Placebo: Civil Servants (Most-Affected Definition Applied)



Notes: The figure illustrates the point estimates of the interaction terms of year dummies for civil servants with the group of most affected workers (based on age and employment duration) and 90% confidence intervals. These estimates are derived from household-level fixed effects regressions that include controls for marriage status, region (East/West Germany), ISCO-08 occupation, number of children, and employment level. The red vertical line marks the time of the reform implementation.
Data Source: German Socioeconomic Panel.

many's current account balance (CA) in partial equilibrium. The GSOEP-based approach takes the estimated reform effects on saving rates from the household fixed-effects regressions in Table 1, maps these effects into aggregate saving flows, ΔS_t , and expresses them as shares of GDP and the CA. The national accounts (NA) approach begins with the published household saving rate, subtracts the estimated Hartz IV effect (in percentage points), converts actual and

counterfactual saving rates into levels using the official denominator, and calculates their difference as ΔS_t . For comparability, both the GSOEP-based and national-accounts calculations are expressed for the population under 65. Together, the GSOEP and NA approaches provide a lower–upper bound: the GSOEP figures likely understate saving at the very top of the distribution and some capital income (due to under-reporting and under-sampling), whereas the NA approach (scaled to under-65s) covers the full household sector (including civil servants, those out-of-the labor force, etc.) for that population share. Our mechanism, therefore, speaks to *flows* of saving out of labor income for a broad group of households, rather than to the level of existing wealth held by the very wealthy households at the top of the wealth distribution.

Using the GSOEP micro estimates, the cumulative change in household saving attributable to the reform amounts to 1.56% of 2014 GDP over 2005–2009; of this, 0.32% is accounted for by the most-affected group (based on its incremental effect and population weight).²⁰ Expressed in current-account terms, the GSOEP-based back-of-the-envelope implies a simple average annual contribution of 6.67% of the observed current-account balance over 2005–2009, of which 1.34% is attributable to the most-affected group. The national-accounts approach yields an upper bound of 8.40% overall and 1.53% for the most-affected group over the same window. Over 2005–2014, the implied shares are very similar (see Table 2). Taken together, these back-of-the-envelope calculations suggest that the reform’s effect on saving explains a non-negligible share of the German current account balance. By construction, however, these static calculations ignore general-equilibrium feedback effects, such as price adjustments, and government balance-sheet channels (e.g., higher employment and lower benefit spending improving the fiscal balance; see Appendix C.3). To explore these channels in a coherent manner, we turn to a structural small open economy model in the next section that links household saving and the current account in general equilibrium.

Table 2: Back-of-the-envelope calculation based on GSOEP and National Accounts

<i>Effects in %</i>	5-year contribution (2005–2009)		10-year contribution (2005–2014)	
	ΔS (cum)/GDP ₂₀₁₄	CA Contr. (avg)	ΔS (cum)/GDP ₂₀₁₄	CA Contr. (avg)
BOE based on GSOEP				
Overall (all HH)	1.56	6.67	3.33	6.22
Most-affected (incr. effect)	0.32	1.34	0.74	1.36
BOE based on National Accounts				
Overall (all HH)	1.97	8.40	4.12	7.72
Most-affected (incr. effect)	0.36	1.53	0.75	1.41

Notes: ΔS Savings (cum)/GDP₂₀₁₄ reports the cumulative change in household savings over the window divided by 2014 GDP in percent. “CA Contr. (avg)” is the simple average across years of the contribution of reform-induced savings to the current-account balance in percent. For both calculations we consider only the share of the population below the age of 65.

Further Empirical Evidence: A comparison of current account movements across European countries shows that, while many Eurozone economies, particularly those in Southern Europe, experienced persistent deficits, Germany’s current account balance stands out in magnitude and

²⁰ For the overall effect, we use the coefficient on the Hartz IV-Dummy in Column (1) of Table 1. For the most-affected group, we multiply the incremental effect in Column (2) by the group’s GSOEP population share.

persistence compared to other Eurozone countries; see Figure 16 in Appendix B.2. Since the introduction of the Euro, current account balances within the Eurozone have increasingly diverged. Notably, Austria, which shares several institutional similarities with Germany, exhibited early dynamics similar to Germany's after the Euro's introduction, but, unlike Germany, did not implement labor market reforms and has not displayed a comparable increase in the current account balance.

Furthermore, we perform time series regressions controlling for macroeconomic variables, including the real effective exchange rate, government debt, and interest rates. The results, depicted in Table 10 of Appendix B.3, indicate a statistically significant increase in Germany's current account balance following the Hartz IV reform. Additionally, rolling-window regressions indicate that the effect was strongest in the first five years post-reform (2005–2009).

3 The Model

We develop a small open economy model with worker heterogeneity, a frictional labor market, and incomplete asset markets. Time is discrete, and the economy consists of a continuum of infinitely-lived consumers who discount the future with a skill-specific factor β^j and derive utility from consumption.²¹ The economy belongs to a currency union, so the risk-free interest rate is taken as exogenous and constant, and there are no aggregate shocks.

Consumers can be employed (E), short-term unemployed (ST), or long-term unemployed (LT), supplying one unit of labor inelastically. Employed workers are either low-skilled (z^L) or high-skilled (z^H), with transitions governed by skill acquisition (probability π^E) and skill depreciation upon job loss. Workers accumulate skills while employed and experience skill deterioration during unemployment (Ljungqvist and Sargent, 2007; Krause and Uhlig, 2012; Hartung et al., 2025).²² We distinguish between short- and long-term unemployment to capture Germany's two-tier benefit system. Short-term benefits are earnings-related and depend on the worker's previous skill-specific wage, while long-term benefits are tied to a fraction of the low-skilled wage.²³ This structure implies differential exposure to benefit exhaustion: formerly high-skilled workers face a larger discrete income drop when transitioning from ST to LT , whereas low-skilled workers face higher exposure through a greater likelihood of unemployment (given their higher job-loss risk).

Wages are set through bargaining between the firm and a labor union for each skill group; for tractability, the union computes the match surplus for a representative member endowed with group-average assets.

The individual states are assets, skill, and labor-market status. This is the minimal heterogeneity needed to implement the German Hartz IV changes in benefit levels and benefit exhaustion, and to translate them into differential exposure and heterogeneous precautionary saving responses,

²¹ A skill-specific discount factor β^j increases wealth dispersion while reducing the need for additional heterogeneity.

²² Our assumption of skill depreciation during unemployment aligns with empirical evidence from German data, as shown in estimated models by Jarosch (2023) and Burdett, Carrillo-Tudela, and Coles (2020), which highlight significant human capital erosion during prolonged unemployment.

²³ This approach, which parallels employment duration heterogeneity, captures the impact of unemployment duration on benefits and skill levels (Hartung et al., 2025; Krause and Uhlig, 2012).

with implications for net foreign asset accumulation.

Several components of our model follow standard approaches from the literature. The heterogeneous agent structure with idiosyncratic risk and precautionary savings follows the incomplete markets framework of [Aiyagari \(1994\)](#). The labor market features a Diamond-Mortensen-Pissarides (DMP) structure ([Pissarides, 2000](#)), in which firms post vacancies and worker-firm matches are formed via a Cobb-Douglas matching function. On the international side, our model builds on the small open economy structure of [Gali and Monacelli \(2005\)](#), in which the domestic economy interacts with the rest of the world through trade and financial flows.

Our framework departs from these benchmark models in several ways. First, unlike [Gali and Monacelli \(2005\)](#), we do not assume perfect risk sharing. Instead, consumers face idiosyncratic, uninsurable employment risk and accumulate precautionary savings, which shape net foreign asset accumulation and trade balance dynamics.²⁴ Second, we introduce a labor market with skill heterogeneity and unemployment duration. While [Krusell et al. \(2010\)](#) incorporate search frictions, we extend their framework by including ex-ante skill differentiation and a detailed unemployment benefit system.

Asset markets are incomplete: agents cannot trade state-contingent claims and therefore cannot insure idiosyncratic labor income risk. They can, however, self-insure by saving in a single non-contingent asset. For expositional convenience, we represent this asset as shares in a competitive investment fund whose shares are held by households and that passively holds the economy's traded claims, including physical capital, firms' equity (dividends), government bonds, and net foreign assets. Because the aggregate environment is deterministic in our setup, equilibrium risk premia are zero, and all traded claims earn a common return.²⁵ Diversification within the bundle has no insurance value at the individual level; all heterogeneity in portfolios arises from differences in the *level* of asset holdings.

There is a positive measure of firms, each hiring one worker and operating in a competitive product market. Firms use capital and labor as inputs, and profits are paid out as dividends d in every period.

3.1 Labor Market Matching

All unemployed workers search on the same labor market. Each period, vacant jobs for workers (V_t) and unemployed workers (U_t) are matched randomly, and the number of matches per period can be described by a Cobb-Douglas matching function:

$$M_t = \chi V_t^{1-\eta} U_t^\eta, \quad (3.1)$$

where χ denotes the matching efficiency and η the matching elasticity. A match is always formed between a firm and a low-skilled worker. The probability that a vacancy is filled in the current

²⁴ Further differences from [Gali and Monacelli \(2005\)](#) are that we abstract from nominal rigidities and explicitly incorporate capital accumulation in our framework.

²⁵ The zero-profit investment fund is therefore a bookkeeping device that implements a single non-contingent saving instrument for households.

period is

$$\lambda_t^f(\theta_t) = \frac{M_t(V_t, U_t)}{V_t} = \chi \theta_t^{-\eta}, \quad (3.2)$$

where $\theta_t = V_t/U_t$ is the ratio of vacancies to unemployed workers and defines labor market tightness. The probability that an unemployed worker finds a job in the current period is

$$\lambda_t^w(\theta_t) = \frac{M_t(V_t, U_t)}{U_t} = \chi \theta_t^{1-\eta}. \quad (3.3)$$

Low-skilled workers can become high-skilled with probability π^E , reflecting skill upgrading during employment. Job destruction is exogenous, occurring with a constant probability that differs by skill level ($s^L > s^H$), where s^L denotes the separation rate for low-skilled workers and s^H is the separation rate for a high-skilled worker.

The law of motion for low-skilled employment N_t^L is given by last period's low-skilled employed workers who did not become high-skilled and new matches:

$$N_{t+1}^L = (1 - s^L)(1 - \pi^E)N_t^L + M_t. \quad (3.4)$$

The next period's pool of high-skilled workers N_t^H consists of high-skilled workers who were not separated from their jobs and low-skilled workers who became high-skilled:

$$N_{t+1}^H = (1 - s^L)\pi^E N_t^L + (1 - s^H)N_t^H. \quad (3.5)$$

The aggregate law of motion for employment is given by

$$N_{t+1} = (1 - s^L)N_t^L + (1 - s^H)N_t^H + M_t. \quad (3.6)$$

The aggregate unemployment rate is given by all workers who are currently not in employment $U_t = 1 - N_t$. If a match gets destroyed, the worker becomes short-term unemployed. In the next period, the worker either finds a job again with probability λ_t^w , enters the pool of long-term unemployment benefit recipients with constant probability π^{LT} , or remains short-term unemployed.²⁶ Once a worker has entered the pool of long-term unemployment benefit recipients, she either finds a job or remains long-term unemployed. The corresponding law of motion of short-term unemployment benefit recipients with skill level $j \in (L, H)$ during their previous employment spell is given by

$$U_{t+1}^{ST,j} = (1 - \pi^{LT})(1 - \lambda_t^w)U_t^{ST,j} + s^j N_t^j, \quad (3.7)$$

where π^{LT} denotes the exogenous probability to switch from short-term into long-term unemployment. Long-term unemployment benefits are independent of the previous skill level. The law of motion for long-term unemployment benefit recipients is:

$$U_{t+1}^{LT} = \pi^{LT}(1 - \lambda_t^w)U_t^{ST} + (1 - \lambda_t^w)U_t^{LT}, \quad (3.8)$$

²⁶ See [Oswald, 1993](#) and [Moyen and Stähler, 2014](#).

where $U_t^{ST} = U_t^{ST,L} + U_t^{ST,H}$. It follows that $U_t = U_t^{ST} + U_t^{LT}$.

3.2 Investment Fund and Asset Structure

A consumer is subject to idiosyncratic, uninsurable employment risk but has access to a single asset supplied by an investment fund. The investment fund collects and aggregates deposits from consumers at no cost.²⁷ Aggregate assets $\bar{a}_t$ are given by

$$\bar{a}_t = \int a f^M(a) da,$$

where $f^M(a)$ denotes the mass of all consumers (employed and unemployed) with wealth level a .

The role of the investment fund is restricted to aggregating different asset classes. It allocates total assets $\bar{a}_t$ across four types: physical capital k_t (valued at $\frac{p_t^H}{P_t}$ in CPI units), government bonds b_t , equity x_t representing claims on firms' profits, and net foreign assets NFA_t . We normalize the total amount of firms' equity to one, thus $x_t = 1$. Following [Krusell et al. \(2010\)](#), we denote the value of a firm by p_t and assume that consumers can only hold claims on aggregate profits rather than profits of individual firms. Consequently, the equity price p_t is determined by the present-value condition

$$p_t = \frac{d_{t+1} + p_{t+1}}{R}, \quad (3.9)$$

where $R = 1 + r - \delta$ denotes the gross risk-free rate. This expression follows from a no-arbitrage condition, which implies the equity return equals the risk-free return implied by the kernel.²⁸ Appendix C.1 shows that, given our assumption of no aggregate risk and an exogenous real interest rate, all traded claims are priced by the same deterministic kernel, so there is no risk–return trade-off across asset classes.

The investment fund operates under the aggregate asset constraint

$$\bar{a}_t = \frac{p_t^H}{P_t} k_t + NFA_t + b_t + p_{t-1}, \quad (3.10)$$

which ensures that the total value of assets equals the sum of capital, government debt, net foreign assets, and equity. We follow the standard ex-dividend timing: p_t denotes the equity price at the end of period t , so the market value of the one unit of equity held at the beginning of period t is p_{t-1} . All assets in this equation are expressed in real terms.²⁹ The law of motion for capital

²⁷ See Appendix C.1 for a formal characterization of the investment fund's problem and the associated no-arbitrage conditions.

²⁸ As highlighted by [Mukoyama \(2013\)](#), equation (3.9) holds at any point where the future path of the aggregate state is known to the consumer. When reforms change the expected profit stream, the value of p_0 must be recomputed accordingly.

²⁹ Physical capital k_t is converted from producer prices to CPI terms, which ensures that all components of total assets $\bar{a}_t$ are measured in consistent units.

defines aggregate investment I_t :

$$\frac{p_t^H}{P_t} I_t = \frac{p_{t+1}^H}{P_{t+1}} k_{t+1} - (1 - \delta) \frac{p_t^H}{P_t} k_t. \quad (3.11)$$

where δ denotes the depreciation rate. Importantly, households do not choose the composition of $\bar{a}_t$ across k_t , b_t , x_t , and NFA_t : they choose only the level of a_t , and the fund then implements the unique composition consistent with (3.10) and the no-arbitrage conditions in Appendix C.1.

3.3 Consumers

All consumers have CRRA preferences over a CES aggregate of domestic and foreign consumption goods,

$$U = \mathbb{E}_0 \sum_{t=0}^{\infty} (\beta^j)^t u(c_t), \quad u(c_t) = \frac{c_t^{1-\sigma}}{1-\sigma}, \quad (3.12)$$

where β^j is the skill-specific discount factor and σ is the coefficient of relative risk aversion. Consumers allocate expenditure between home and foreign goods according to a CES aggregator with home-bias parameter $\gamma \in (0, 1)$ and curvature parameter $\eta^C \in (0, 1)$,

$$c_t = \left(\gamma^{1-\eta^C} c_{H,t}^{\eta^C} + (1-\gamma)^{1-\eta^C} c_{F,t}^{\eta^C} \right)^{\frac{1}{\eta^C}}. \quad (3.13)$$

where $c_{H,t}$ denotes consumption of home-produced goods and $c_{F,t}$ denotes consumption of foreign-produced goods. The implied elasticity of substitution between home and foreign goods is $\sigma^C = \frac{1}{1-\eta^C}$.

Employed consumers receive after-tax labor income $(1 - \tau^w)w_t$, where w_t denotes the real wage (to be defined later). Unemployed agents receive unemployment benefits that depend on unemployment duration. Short-term benefits are earnings-related and equal $\kappa_t^{ST,j} = b^{ST}(1 - \tau^w)w_t^j$, while long-term benefits equal $\kappa_t^{LT} = b^{LT}(1 - \tau^w)w_t^L$, with $b^{ST} > b^{LT}$. Each period, consumers take prices and wages as given and choose consumption and asset holdings.

Aggregate State: The aggregate state Z_t represents the distribution of agents over individual states—assets a , employment status e , and skill type $j \in (L, H)$. Z_t is a probability measure on the state space $\mathcal{A} \times \mathcal{E} \times \mathcal{J}$, defined over the Borel σ -algebra $\Sigma_Z = \mathcal{B}(\mathcal{A}) \otimes \mathcal{P}(\mathcal{E}) \otimes \mathcal{P}(\mathcal{J})$. The evolution of Z_t is deterministic and follows the law of motion:

$$Z_{t+1} = \Gamma(Z_t),$$

where Γ is a deterministic transition function derived from agents' optimal policies and the individual transition probabilities.

Low-skilled Employed Consumers: The recursive formulation of the low-skilled employed consumer's problem is

$$V_t^{E,L}(a_t, Z_t) = \max_{\{c_t^{E,L}, a_{t+1}\}} \left\{ u(c_t^{E,L}) + \beta^L \left[(1-s^L)(1-\pi^E) V_{t+1}^{E,L}(a_{t+1}, Z_{t+1}) \right. \right. \\ \left. \left. + (1-s^L)\pi^E V_{t+1}^{E,H}(a_{t+1}, Z_{t+1}) + s^L V_{t+1}^{ST,L}(a_{t+1}, Z_{t+1}) \right] \right\}, \quad (3.14)$$

subject to

$$c_t^{E,L} + a_{t+1} = (1+r-\delta)a_t + (1-\tau^w)w_t^L(Z_t) - \tau_t(Z_t), \\ a_{t+1} \geq \underline{a}.$$

where $u(\cdot)$ denotes the increasing and concave utility function. Let the decision rule for a_{t+1} for low-skilled employed workers be $a_{t+1} = \varphi_t^{E,L}(a)$. The consumption of employed workers is denoted $c_t^{E,L}$, $(1+r-\delta)$ represents the interest rate net of depreciation δ on assets a_t , τ^w denotes a labor-income tax on the wage of low-skilled workers, w_t^L , τ_t is a lump-sum tax (or transfer), and $\underline{a}$ represents the borrowing constraint.

High-skilled Employed Consumers: The recursive formulation of the high-skilled employed consumer's problem is

$$V_t^{E,H}(a_t, Z_t) = \max_{\{c_t^{E,H}, a_{t+1}\}} \left\{ u(c_t^{E,H}) + \beta^H \left[(1-s^H) V_{t+1}^{E,H}(a_{t+1}, Z_{t+1}) \right. \right. \\ \left. \left. + s^H V_{t+1}^{ST,H}(a_{t+1}, Z_{t+1}) \right] \right\}, \quad (3.15)$$

subject to

$$c_t^{E,H} + a_{t+1} = (1+r-\delta)a_t + (1-\tau^w)w_t^H(Z_t) - \tau_t(Z_t), \\ a_{t+1} \geq \underline{a}.$$

where the decision rule for a_{t+1} for high-skilled employed workers be $a_{t+1} = \varphi_t^{E,H}(a)$.

Short-term Unemployed Consumers: A short-term unemployed consumer of skill group $j \in \{L, H\}$ receives unemployment benefits $\kappa_t^{ST,j}$ based on the short-term replacement rate b^{ST} and the previous skill-specific wage w_t^j , such that $\kappa_t^{ST,j} = b^{ST}(1-\tau^w)w_t^j$.³⁰ With exogenous probability π^{LT} , the worker becomes long-term unemployed if not re-employed in the next period. The recursive problem is

$$V_t^{ST,j}(a_t, Z_t) = \max_{\{c_t^{ST,j}, a_{t+1}\}} \left\{ u(c_t^{ST,j}) + \beta^j \left[\lambda_t^w(Z_t) V_{t+1}^{E,L}(a_{t+1}, Z_{t+1}) \right. \right. \\ \left. \left. + (1-\lambda_t^w(Z_t)) \left[(1-\pi^{LT}) V_{t+1}^{ST,j}(a_{t+1}, Z_{t+1}) + \pi^{LT} V_{t+1}^{LT}(a_{t+1}, Z_{t+1}) \right] \right] \right\}, \quad (3.16)$$

³⁰ In equilibrium, $w_t^H > w_t^L$, so $\kappa_t^{ST,H} > \kappa_t^{ST,L}$.

subject to

$$\begin{aligned} c_t^{ST,j} + a_{t+1} &= (1 + r - \delta)a_t + \kappa_t^{ST,j}(Z_t) - \tau_t(Z_t), \\ a_{t+1} &\geq \underline{a}. \end{aligned}$$

Let the decision rule for a_{t+1} for short-term unemployed workers be $a_{t+1} = \varphi_t^{ST,j}(a)$.

Long-term Unemployed Consumers: A long-term unemployed consumer receives less generous unemployment benefits $\kappa_t^{LT} = b^{LT}(1 - \tau^w)w_t^L$, based on the lower replacement rate b^{LT} and the equilibrium wage of low-skilled workers.³¹ They either find a job with probability λ_t^w in the next period or remain long-term unemployed. Their problem is:

$$\begin{aligned} V_t^{LT}(a_t, Z_t) = \max_{\{c_t^{LT}, a_{t+1}\}} & \left\{ u(c_t^{LT}) + \beta^L \left[\lambda_t^w(Z_t) V_{t+1}^{E,L}(a_{t+1}, Z_{t+1}) \right. \right. \\ & \left. \left. + (1 - \lambda_t^w(Z_t)) V_{t+1}^{LT}(a_{t+1}, Z_{t+1}) \right] \right\}, \end{aligned} \quad (3.17)$$

subject to

$$\begin{aligned} c_t^{LT} + a_{t+1} &= (1 + r - \delta)a_t + \kappa_t^{LT}(Z_t) - \tau_t(Z_t), \\ a_{t+1} &\geq \underline{a}, \end{aligned}$$

where $a_{t+1} = \varphi_t^{LT}(a)$ denotes the decision rule for long-term unemployed workers.

3.4 Firms

Firms produce output with two inputs: capital and labor. They hire workers on a frictional labor market and rent capital. A matched worker-firm pair produces output according to

$$y_t^j = z^j (\tilde{k}_t^j)^\alpha. \quad (3.18)$$

where z^j is a skill-specific total factor productivity parameter, $\tilde{k}_t^j$ is the amount of capital per worker and defined as the ratio of aggregate capital per employed worker $\tilde{k}_t^j = \frac{k_t^j}{N_t^j}$. The aggregate output is then given by $Y_t = y_t^L N_t^L + y_t^H N_t^H$. The real per-period firm profit for a worker is defined as

$$\Pi_t^j = \frac{p_t^H}{P_t} y_t^j - \frac{p_t^H}{P_t} r \tilde{k}_t^j - w_t^j. \quad (3.19)$$

The dividends paid out are given by aggregate profits (per-worker firm profits Π_t^j times the number of employed workers N_t^j for low-skilled and high-skilled worker-firm pairs respectively) mi-

³¹ We assume that low-skilled workers dominate the pool of long-term unemployment benefit recipients.

nus vacancy posting costs:

$$d_t = \Pi_t^L N_t^L + \Pi_t^H N_t^H - \kappa^v V_t, \quad (3.20)$$

where κ^v denotes real vacancy posting costs. Firms maximize the present value of the discounted stream of dividends. Given r , capital demand satisfies

$$r = \alpha z^j (\tilde{k}_t^j)^{\alpha-1}. \quad (3.21)$$

Note that the rental rate on capital r is fixed on the international market in the setting of a small open economy.

The marginal value of an additional low-skilled worker to the firm is

$$\begin{aligned} J_t^L(Z_t) = & \frac{p_t^H(Z_t)}{P_t(Z_t)} z^L (\tilde{k}_t^L)^\alpha - \frac{p_t^H(Z_t)}{P_t(Z_t)} r \tilde{k}_t^L - w_t^L(Z_t) \\ & + q \left[(1-s^L)(1-\pi^E) J_{t+1}^L(Z_{t+1}) + (1-s^L)\pi^E J_{t+1}^H(Z_{t+1}) + s^L \psi_{t+1}^{V,L}(Z_{t+1}) \right], \end{aligned} \quad (3.22)$$

which takes into account that with probability π^E the worker may become high-skilled in the next period. Additionally, with probability s^L , the match is destroyed, in which case the firm receives the continuation value of a vacant position, $\psi_{t+1}^{V,L}$. The term q represents the inverse of the gross real interest rate, $q = \frac{1}{1+r-\delta}$. The marginal value of a high-skilled worker to the firm is

$$\begin{aligned} J_t^H(Z_t) = & \frac{p_t^H(Z_t)}{P_t(Z_t)} z^H (\tilde{k}_t^H)^\alpha - \frac{p_t^H(Z_t)}{P_t(Z_t)} r \tilde{k}_t^H - w_t^H(Z_t) \\ & + q \left[(1-s^H) J_{t+1}^H(Z_{t+1}) + s^H \psi_{t+1}^{V,H}(Z_{t+1}) \right]. \end{aligned} \quad (3.23)$$

The value of a vacancy, $\psi_t^{V,j}$, is

$$\psi_t^{V,j}(Z_t) = -\kappa^v + q \left[(1-\lambda_t^f) \psi_{t+1}^{V,j}(Z_{t+1}) + \lambda_t^f J_{t+1}^j(Z_{t+1}) \right], \quad (3.24)$$

where in equilibrium, firms will post vacancies until $\psi_t^{V,j} = 0$ holds.

The job-creation condition assuming free market entry is given by

$$\frac{\kappa^v}{\lambda_t^f} = q \left[J_{t+1}^L(Z_{t+1}) \right]. \quad (3.25)$$

3.5 Wage Determination

We assume that the surplus generated by a match between workers and firms is shared according to a surplus-sharing rule that ensures equal gains for both parties. Atomistic labor unions, representing the average worker, negotiate wages with firms.

This approach is empirically grounded and offers tractability for our analysis. Empirically, it reflects the significant role of sectoral agreements in the German labor market during the period of the reform. Before the reform, collective bargaining was the predominant wage-setting mecha-

nism in Germany. According to the IAB Establishment Panel, 68% of employees in western Germany were covered by collective agreements in 2004, with sectoral agreements playing the largest role, while company-level agreements covered only 2% of establishments and 7% of employees (Elguth and Kohaut, 2005). Technically, modeling wage determination through unions that bargain on behalf of the average worker simplifies the complex dynamics that arise in search-and-matching frameworks with heterogeneous agents and asset accumulation, especially when calculating dynamic transition paths.³²

The union's objective is to maximize the surplus of employment over unemployment for the average worker, reflecting a central motive for members' welfare and aiming to secure outcomes representative of workers' interests.³³ This approach accounts for the average consumption and asset levels of its members in the wage bargaining process.³⁴

The value of an employed worker in skill group j is given by the value function $V^{E,j}(a_t, Z_t)$. For the average worker in this group, we evaluate the function at the group-specific mean asset level, denoted by $\tilde{a}_t$, and define this as:

$$\mathcal{W}_t^{E,j}(Z_t) = V^{E,j}(\tilde{a}_t, Z_t).$$

Similarly, the values of short-term unemployed and long-term unemployed workers in skill group j are given by the corresponding value functions, $V^{ST,j}(a_t, Z_t)$ and $V^{LT}(a_t, Z_t)$, respectively. Evaluating these functions at the average values of the individual states, we define the corresponding values as:

$$\mathcal{W}_t^{ST,j}(Z_t) = V^{ST,j}(\tilde{a}_t, Z_t), \quad \mathcal{W}_t^{LT}(Z_t) = V^{LT}(\tilde{a}_t, Z_t).$$

Evaluating value functions at the group-specific mean asset level is a tractable way to let wages depend on workers' outside options without introducing the full cross-sectional asset distribution as an additional state variable in wage setting; it preserves the economic logic of outside option wage pressure while keeping the bargaining problem finite-dimensional.

The fallback position of a union member in union j is the payoff during unemployment, which can be short-term or long-term unemployment (weighted by the corresponding share).

We assume that the union's surplus is given by its average members' difference between the values of being employed and the fallback position of unemployment.³⁵

$$\tilde{\mathcal{W}}_t^j(Z_t) = \mathcal{W}_t^{E,j}(Z_t) - \left(\mathcal{W}_t^{ST,j}(Z_t) \frac{U_t^{ST}}{U_t^{ST} + U_t^{LT}} + \mathcal{W}_t^{LT}(Z_t) \frac{U_t^{LT}}{U_t^{ST} + U_t^{LT}} \right). \quad (3.26)$$

³² Bilateral bargaining between workers and firms leads to a wage schedule that rises with wealth. Solving the model thus involves determining this endogenous wage schedule, adding another layer of complexity as it represents an infinite-dimensional object.

³³ Using the median worker instead would amplify the net foreign asset effects due to lower initial assets and higher savings. Consequently, modeling the average worker provides a more conservative approach.

³⁴ Modeling wage determination through a labor union is a standard approach in heterogeneous agent models (see Auclert, Rognlie, and Straub, 2024; de Ferra et al., 2020; Bayer et al., 2024 who abstract from search and matching frictions).

³⁵ Note that this setup is not efficient as long-term unemployed would have a lower reservation wage.

The firm's surplus is given by the marginal value of one additional worker to a firm, $J_t^i(Z_t)$, see equations 3.22 and 3.23.

Bargaining Solution

To solve for the bargained wage, we assume Egalitarian bargaining (see [Kalai, 1977](#)), which in our context ensures equal gains for both parties upon reaching an agreement.³⁶ The wage is implicitly characterized by the following surplus-sharing rule:

$$\tilde{w}_t^j(Z_t) = \frac{\zeta}{(1-\zeta)} J_t^j(Z_t). \quad (3.27)$$

By adopting Egalitarian bargaining, our model remains tractable, eliminates the endogenous wealth-wage feedback loop that would arise under Nash bargaining (see [Bardóczy, 2017](#) for a discussion), and ensures a stable wage structure. This makes it particularly suitable for environments with precautionary savings and risk-averse workers.³⁷ Moreover, controlled experiments indicate that real-world bargaining behavior frequently aligns more closely with the Egalitarian solution than with the Nash solution (see [Bruce and Clark, 2012](#)).

3.6 Fiscal Authority

The fiscal authority finances unemployment benefits for short and long-term unemployed workers as well as interest payments on outstanding government debt with a lump sum tax τ , a labor-income tax τ^w , and by issuing new government bonds b_{t+1} :

$$\kappa_t^{ST,L} U_t^{ST,L} + \kappa_t^{ST,H} U_t^{ST,H} + \kappa_t^{LT} U_t^{LT} + (1+r-\delta)b_t = \tau^w(w_t^L N_t^L + w_t^H N_t^H) + \tau_t + b_{t+1}. \quad (3.28)$$

To ensure stationarity of government debt (see [Schmitt-Grohe and Uribe, 2007](#)), the lump-sum tax follows the following tax rule:

$$\tau_t - \bar{\tau} = \rho^\tau (\tau_{t-1} - \bar{\tau}) + (1 - \rho^\tau) \chi^b (b_{t-1} - \omega^B Y_t). \quad (3.29)$$

where $\bar{\tau}$ denotes the steady state level of the lump-sum tax, ρ^τ is a smoothing parameter, and χ^b determines the elasticity of the tax rate to deviations of government debt from the target debt level $\omega^B Y_t$, where ω^B denotes the target debt-to-output ratio.

3.7 Rest of the Currency Zone

The small open economy is linked to the rest of the currency zone through trade in consumption goods and international assets. Let p_t^H and p_t^F denote the prices of the home and foreign goods,

³⁶ Egalitarian bargaining satisfies the axioms Pareto-optimality, symmetry, and strong monotonicity. However, as opposed to the Nash bargaining solution, the solution is not scaled by the wage's marginal utility and, therefore, is not scale invariant (see [l'Haridon, Malherbet, and Pérez-Duarte, 2013](#) for a discussion).

³⁷ As shown in [l'Haridon et al. \(2013\)](#), in a model without precautionary savings, Nash and Egalitarian bargaining yield equivalent outcomes when workers are risk-neutral. However, under risk aversion, this equivalence breaks, though the quantitative differences remain small.

respectively. We define the real exchange rate as the relative price of foreign to home goods, $\text{rer}_t = \frac{p_t^F}{p_t^H}$, and the terms of trade as its inverse, $\text{tot}_t = \frac{p_t^H}{p_t^F}$.

The consumer price index is given by

$$P_t = \left[\gamma (p_t^H)^{-\eta^C / (1-\eta^C)} + (1-\gamma) (p_t^F)^{-\eta^C / (1-\eta^C)} \right]^{-(1-\eta^C) / \eta^C}. \quad (3.30)$$

The demand for consumption goods of the rest of the currency union produced in home c_H^* is a function of the terms of trade:

$$c_{H,t}^* = \text{tot}_t^{-\theta^F} X \quad (3.31)$$

where θ^F denotes the demand elasticity of the foreign population for changes in the relative price between goods produced at home and abroad and X is an exogenous demand shifter capturing the size of the foreign market (see [de Ferra et al., 2020](#)). Demand for home and foreign consumption goods can be expressed as

$$c_{H,t} = \left(\frac{\gamma}{(1-\gamma)} \right) \text{rer}_t^{\frac{1}{1-\eta^C}} c_{F,t}. \quad (3.32)$$

Asset market clearing implies that total assets in the home economy, $\bar{a}_t$, have to equal government debt plus net foreign assets, capital, and equities. Hence, the aggregate asset constraint defined in Equation 3.10 must hold. A country's net foreign asset position is defined as last period's assets plus current net exports, NX_t ,

$$\text{NFA}_{t+1} = (1 + r - \delta) \text{NFA}_t + NX_t, \quad (3.33)$$

where the difference between exports and imports gives net exports:

$$NX_t = EX_t - IM_t = \frac{p_t^H}{P_t} c_{H,t}^* - \frac{p_t^F}{P_t} c_{F,t}. \quad (3.34)$$

The current account is given by the change in the net foreign asset position, $\text{CA}_t = \text{NFA}_{t+1} - \text{NFA}_t$.

3.8 Market Clearing

Total resources in the domestic economy equal aggregate domestic consumption, investment, and resource costs arising from vacancy posting. Hence, domestic good market clearing implies:

$$Y_t = c_{H,t} + c_{H,t}^* + I_t + \frac{P_t}{p_t^H} \kappa^v V_t. \quad (3.35)$$

Appendix C.2 contains a detailed derivation of the resource constraint.

3.9 Equilibrium

Given the exogenous net risk-free rate $r - \delta$, the recursive competitive equilibrium of the small open economy consists of prices $\{w_t^L, w_t^H, P_t, p_t^H, p_t^F, p_t\}$, value functions

$\{V_t^{E,L}(a_t, Z_t), V_t^{E,H}(a_t, Z_t), V_t^{ST,L}(a_t, Z_t), V_t^{ST,H}(a_t, Z_t), V_t^{LT}(a_t, Z_t), J_t^L(Z_t), J_t^H(Z_t), \gamma_t^{V,j}\}$, consumers' decision rules for asset holdings $\{\varphi_t^{E,j}(a), \varphi_t^{ST,j}(a), \varphi_t^{LT}(a)\}$, a law of motion for the distribution Z_t , and aggregate allocations, such that:

1. Consumers solve their optimization problems given by Equations (3.14) to (3.17), subject to their budget constraints.
2. Firms optimize, taking prices and the aggregate state Z_t as given. The number of vacancies V_t is determined by the free-entry condition, implying that the value of posting a vacancy satisfies $\gamma_t^{V,j} = 0$.
3. The law of motion for the distribution Z_t is consistent with individual decision rules.
4. The job-finding rate λ_t^w and the job-filling rate λ_t^f depend on market tightness θ_t via the matching function:

$$\lambda_t^w = \lambda^w(\theta_t), \quad \lambda_t^f = \lambda^f(\theta_t), \quad \theta_t = \frac{V_t}{U_t}.$$

Employment evolves according to the law of motion (3.6).

5. The numbers of short-term unemployed and long-term unemployed follow (3.7) and (3.8).
6. Wages are determined by the surplus splitting rule in Equation (3.27).
7. The government budget constraint (3.28) holds, and the lump-sum tax rate is determined by the tax rule (3.29).
8. Asset markets clear, all assets yield the same risk-free return in equilibrium (no-arbitrage) and the asset market clearing condition (3.10) is satisfied.
9. The aggregate resource constraint (3.35) is satisfied.

4 Calibration and Reform Implementation

In this section, we first describe the calibration of our model and then discuss the implementation of the German labor market reform.

4.1 Calibration

We calibrate the model to a quarterly frequency. Table 3 presents our baseline parameter choices, while Table 4 lists our calibration targets. We assume permanent discount factor heterogeneity with $\beta \pm \Delta$ to obtain a realistic level of wealth inequality, where high-skilled workers are more patient (with a discount factor $\beta^H = \beta + \Delta$) and low-skilled workers are less patient (with a discount factor $\beta^L = \beta - \Delta$).³⁸ We set the (exogenous) net risk-free interest rate on the household asset, $r - \delta$, to 1.5% per quarter and choose the discount factor β and Δ to match the share of wealth held by the bottom 50% of the German wealth distribution, which is 2.1% based on the

³⁸ This is a standard feature in the literature to obtain a realistic distribution of wealth, see [Carroll, Slacalek, Tokunaka, and White \(2017\)](#) and [Auclert et al. \(2021\)](#).

2002 wave of the German Socioeconomic Panel, resulting in a model-implied Gini coefficient of 0.71. Consequently, we set the discount factor to $\beta = 0.981$ with $\Delta = 0.006$, resulting in a firm discount factor of $q = 0.985$. We set the parameter of relative risk aversion, σ , to 2, which is a standard value in the literature.

We normalize low-skilled productivity to $z^L = 1$ and set high-skilled productivity to $z^H = 1.559$, based on the ratio of average wages of high- and low-skilled workers in the GSOEP, which we use as a proxy for relative productivity.³⁹ Following [Bayer et al. \(2024\)](#), we set the capital share to $\alpha = 0.32$ and assume a quarterly depreciation rate of $\delta = 0.025$.

Regarding the labor market, we set the bargaining power of the union, ζ , to 0.5 and the elasticity of matches with respect to unemployment to 0.5, both are standard values in the literature. We set the replacement rate for short-term unemployed, b^{ST} , to 0.67, and for long-term unemployed, b^{LT} , to 0.57, corresponding to the legal values (for individuals with children) before the reform. The probability of transitioning to long-term unemployment benefits is 0.125, resulting in an expected duration of short-term unemployment benefits of 8 quarters, as was typical before the reform. The probability of becoming a high-skilled worker is set to target a low-skilled worker share of 68% ([OECD, 2004](#)), resulting in a probability π^E of 1/230. Separation rates for high-skilled and low-skilled workers are set according to [Hartung, Jung, and Kuhn \(2022\)](#), with 0.89% for high-skilled workers and 4.14% for low-skilled workers based on IAB data.⁴⁰ We target a steady state unemployment rate of 12%, reflecting the prevailing unemployment rate at the time of the reform, and a job-filling rate λ^f of 0.7, according to empirical evidence for the Euro area as reported in [Christoffel, Kuester, and Linzert \(2009\)](#). These targets endogenously determine the matching efficiency χ and vacancy posting costs κ^v in the steady state.

Regarding the open-economy parameters, we set the home-bias parameter for domestic goods to $\gamma = 0.66$, a standard value in the literature. The CES curvature parameter governing substitutability between home and foreign goods is set to $\eta^C = 0.744$. Given our parametrization, this implies an elasticity of substitution $\sigma^C \equiv \frac{1}{1-\eta^C}$ of around 3.9, which lies in the range typically used in quantitative trade and open-economy models. Following [Feenstra, Luck, Obstfeld, and Russ \(2018\)](#), we set the elasticity of foreign demand with respect to the terms of trade to $\theta^F = 3$ (see also [de Ferra et al., 2020](#)).

For the policy parameters, we set the autocorrelation of the lump sum tax rate to 0.95, reflecting the delayed reduction in German social security contributions for unemployment benefits after the reform (see Figure 19 in the Appendix). To ensure stationarity of government debt, we set the response to deviations in government debt from its target at $\chi^b = 0.02$ (see [Kirsanova and Wren-Lewis, 2012](#)).⁴¹ Lastly, consistent with Germany's long-term debt-to-output ratio of 60%, we set $\omega^B = 0.6$.

In the initial steady state, we normalize all prices, the real exchange rate, and the terms of trade to one and assume zero inflation (Table 4). We also normalize the pre-reform net foreign asset position to zero, so the steady state current account is zero given $CA_t = NFA_{t+1} - NFA_t$. Finally,

³⁹ We use years of education as a proxy for skills, classifying individuals with more than 13 years of education as high-skilled workers and those with 13 years or fewer as low-skilled workers.

⁴⁰ Quarterly rates computed from monthly hazards in [Hartung et al., 2022](#) (working paper); published version [Hartung et al. \(2025\)](#).

⁴¹ We provide robustness checks regarding the values of these two parameters in Section 5.2.

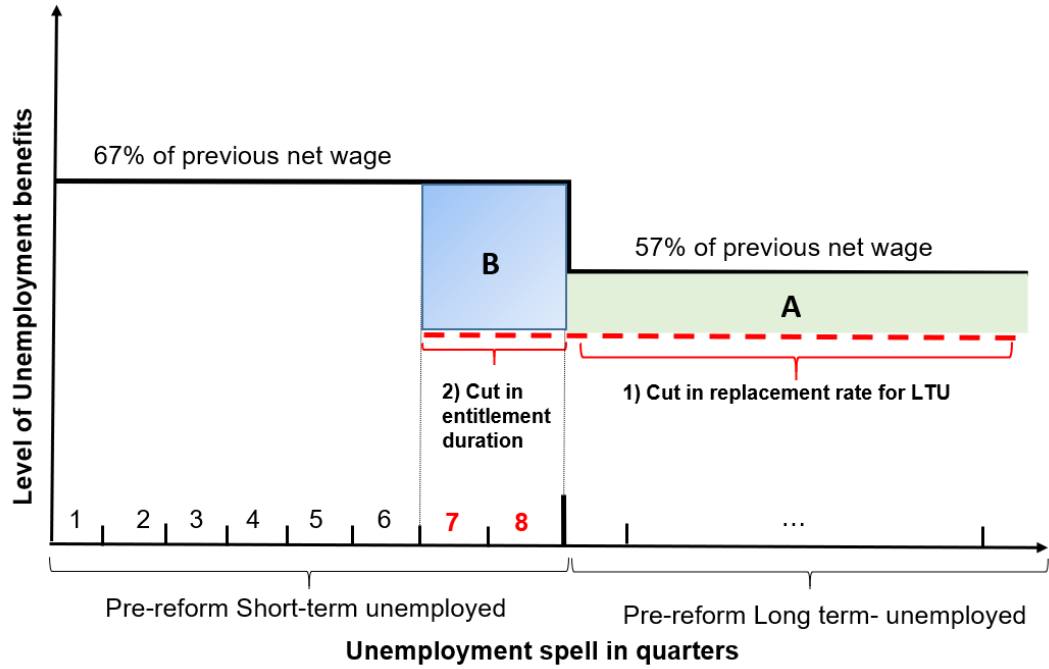
Table 3: Calibration and Targets

Parameters	Symbol	Value	Source/Target
Preferences			
Discount factor	β	0.981	Net interest rate of 1.5%
Discount factor spread	Δ	0.006	Wealth share of bottom 50% (2.1%)
Risk aversion	σ	2.000	Standard Value
Technology			
Total Factor Productivity, Low Skilled	z^L	1.000	Normalization
Total Factor Productivity, High Skilled	z^H	1.559	Relative mean wages of high-vs. low-skilled (GSOEP)
Capital share	α	0.320	Bayer et al. (2024)
Capital depreciation rate	δ	0.025	Standard value
Labor Market			
Workers' bargaining power	ζ	0.500	Standard Value
Matching elasticity	η	0.500	Standard Value
Replacement rate for STU	b^{ST}	0.670	Institutional value
Replacement rate for LTU	b^{LT}	0.570	Institutional value
Probability to switch to LTU	π^{LT}	0.125	Institutional value
Probability to become high skilled	π^E	1/230	Share of low-skilled employed workers
Separation Rate, High-Skilled Workers	s^H	0.89%	Hartung et al. (2022)
Separation Rate, Low-Skilled Workers	s^L	4.14%	Hartung et al. (2022)
Matching efficiency	χ	0.399	Job-filling rate of 0.7 (Christoffel et al., 2009)
Vacancy posting costs	κ^v	0.241	Unemployment rate of 12%
Open Economy			
Home Bias	γ	0.660	Standard value
CES curvature parameter	η^C	0.744	Standard value
Elasticity of foreign demand to rer	θ^F	3.000	Feenstra et al. (2018)
Policy			
Smoothing parameter of fiscal rule	ρ^τ	0.950	Gradual adjustment of German SSC after reform
Tax rate elasticity to debt deviations	χ^b	0.020	Kirsanova and Wren-Lewis (2012)
Long-run government debt-output ratio	ω^B	0.600	German Institutional Target

Table 4: Targets

	Symbol	Value	Source/Target
Targets			
Price of home goods	p_H	1	Normalization
Price of foreign goods	p_F	1	Normalization
Consumer Price Index (CPI)	P	1	Normalization
Real Exchange Rate	rer	1	Normalization
Terms of Trade	tot	1	Normalization
Net Foreign Assets/Y		0.00%	Normalization
Unemployment rate	U	12.0%	Value end of 2004
Job-filling rate	λ^f	0.7	Euro area evidence (see Christoffel et al., 2009)
Share of low-skilled employed workers		68.00%	OECD (2004)
Wealth Share of Bottom 50%		2.1%	GSOEP
Net interest rate	$r - \delta$	1.50%	Historical average for Germany (2001–2004)
Aggregate private savings rate increase		1.0 pp	See Section 2

Figure 5: Reform Implementation (schematic plot)



we choose the size of the replacement-rate reduction so that the model generates an increase in the aggregate private savings rate of 1.0 percentage point within the first ten years, consistent with our empirical evidence in Section 2 and as discussed below.

4.2 Reform Implementation

The Hartz IV reform was undertaken in two steps. First, in 2005, the replacement rate for long-term unemployment benefits was reduced, fixed, and independent of prior earnings. One year later, from 2006 onward, the entitlement duration for receiving short-term unemployment benefits was reduced (see Appendix A for details). On average, the entitlement duration was cut by six months.⁴² Figure 5 schematically plots how we incorporate these changes into our model.

The precise economy-wide reduction in the replacement rate due to the Hartz IV reform remains debated. [Launov and Wälde \(2013\)](#) estimated a 7% decline, leading to a 0.1 percentage point drop in unemployment, while [Krebs and Scheffel \(2013\)](#) and [Krause and Uhlig \(2012\)](#) assumed reductions of 20% and up to 6%, implying larger effects (see [Hochmuth et al., 2021](#) for a detailed discussion). We set the reduction in the replacement rate to match the empirically estimated post-reform increase in private savings rates. This is done by simulating a panel of 50,000 individuals in our model and applying the same regression as in the GSOEP data. Based on this approach, we set the reduction in b^{LT} to 26%, a value within the plausible range and close to the average reported by [OECD \(2019\)](#). The simulation spans 40 quarters (10 years) before and

⁴² The entitlement cut varied by age group and was strongest for elderly workers. For them, the entitlement duration was reduced from a maximum of 32 months to a maximum of 18 months.

after the reform, aligning with the time horizon of the shift dummy in Table 1. Using the same macroeconomic control, we ensure a consistent estimation of the partial effect on aggregate savings while accounting for general equilibrium effects. Since the reduction in the replacement rate is chosen to match the observed savings response, we obtain the same coefficient on the Hartz IV-Dummy by construction (see Table 11 in Appendix D.1).

Furthermore, we model the cut in entitlement duration by increasing the probability of transitioning from short-term to long-term unemployment benefits (π^{LT}) from 1/8 to 1/6, a 33.33% increase. Consequently, the maximum duration of short-term unemployment benefits decreases from 8 to 6 quarters, aligning with the observed reduction for an average worker.

Regarding the simulation design, we assume the economy was in its initial steady state before the policy change in 2005, with no future shocks occurring afterward. As a robustness check, we account for anticipatory effects starting one year before the reform's implementation. We solve for the transition paths between the pre-reform and post-reform steady states under perfect foresight.

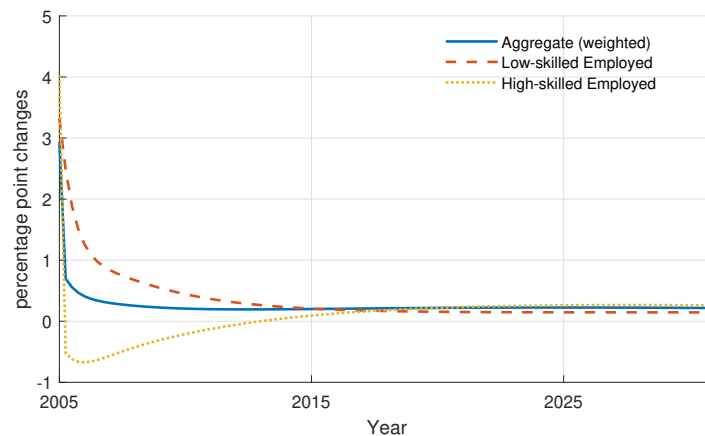
5 Aggregate Effects of Labor Market Reform

In this section, we discuss the aggregate results of our model simulations before we investigate the role of heterogeneity in more detail in the next section. First, we describe the dynamics of our model economy along the transition path. We then calculate the reform's contribution to German and international macroeconomic time series and provide robustness checks.

5.1 Transitional Dynamics

This subsection discusses the transitional dynamics after the Hartz IV reform in Germany. We provide a comparison of pre- and post-reform steady states in Appendix D.2.

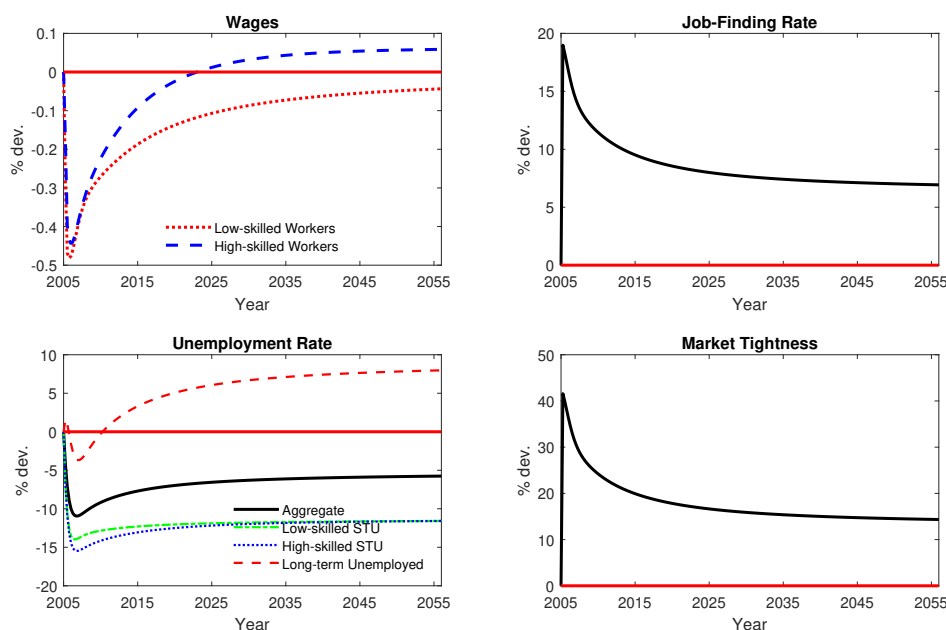
Figure 6: Effect on Saving Rates



Notes: Model responses of saving rates of all workers (weighted with their corresponding shares), low-skilled employed, and high-skilled employed workers in percent deviations from the pre-reform steady state.

Saving Rates: Figure 6 presents the model-based responses of savings rates at the aggregate level and for low-skilled and high-skilled workers separately. The savings rate for low-skilled workers increases sharply before gradually declining to a new steady state. Several factors drive this stronger response among low-skilled workers. First, they face a higher (exogenous) job separation rate and are, therefore, more likely to fall into unemployment in response to the reform, which increases their need for precautionary savings. Second, a larger share of their income comes from labor earnings, making them more sensitive to changes in labor market conditions. Importantly, in steady state, low-skilled workers tend to have lower levels of self-insurance, as they are more likely to be near the borrowing constraint. This amplifies their savings response, as they want to increase their asset holdings. The stronger response of low-skilled workers aligns well with the more pronounced reaction of the most affected workers in our empirical analysis (see Table 1), though the former serves only as a rough proxy for this group. In contrast, high-skilled workers initially experience a decline in their savings rate. This is due to improved labor market conditions, which reduce their need for precautionary savings. Additionally, their pre-reform wealth levels provide greater self-insurance, thereby muting their overall response.

Figure 7: Reform Effects on Labor Market Outcomes



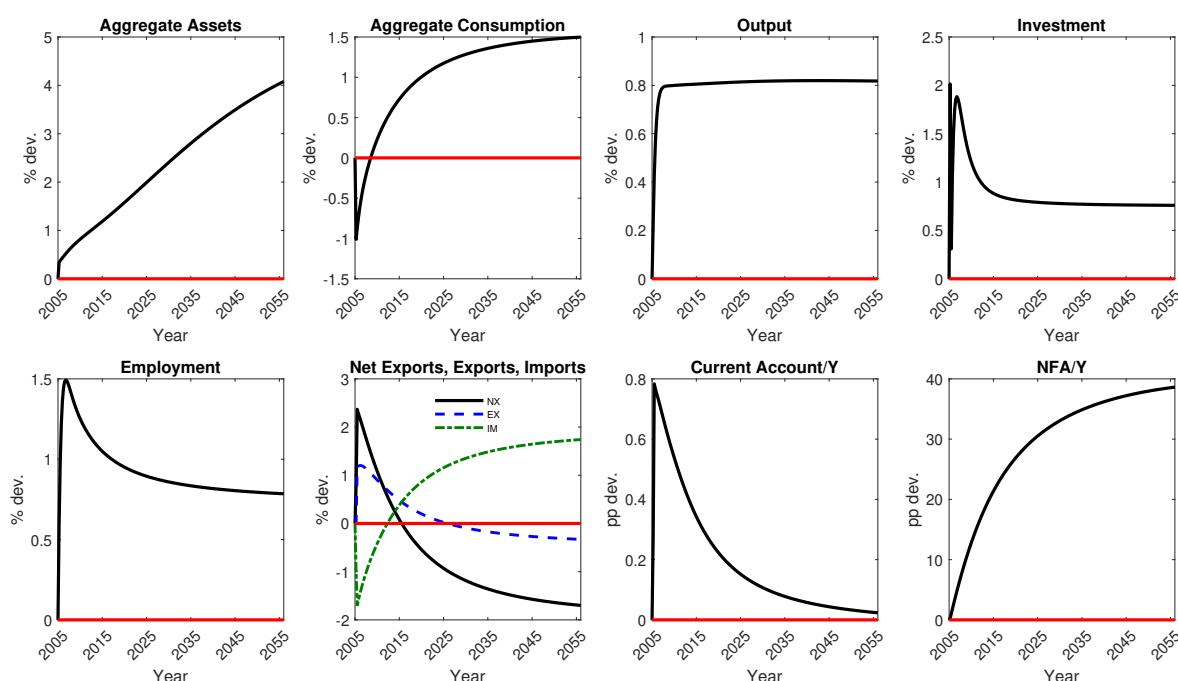
Notes: Model responses of labor market variables in percent deviations from the pre-reform steady state.

Labor Market: Figure 7 shows the reform effects on various labor market outcomes. The reduction in the generosity of the unemployment benefit scheme leads to an initial decline in wages as workers' bargaining positions worsen due to the reduced fall-back utility. Gradually, wages recover over time and are higher for high-skilled workers in the post-reform equilibrium.⁴³ Lower wages increase the marginal value of a worker to firms. As a result, they post more vacancies and the job-finding rate increases. Aggregate unemployment decreases by more than 10% in the

⁴³ Our short-run wage effect is quantitatively in line with causal microeconomic evidence of [Price \(2018\)](#), who rules out sizeable adverse wage effects caused by the Hartz IV reform.

short run. However, the reduction is less pronounced for long-term unemployed workers, as the reform accelerates their transition into long-term unemployment. This dynamic partially offsets the increased job-finding rate, leading to a modest increase in unemployment among the long-term unemployed in the medium and long run. This trend aligns with the rather stagnant number of long-term unemployed individuals in Germany after the implementation of the reform, as discussed by [Spermann \(2015\)](#).

Figure 8: Aggregate Reform Effects



Notes: Model responses of aggregate variables in percent deviations (percentage point deviations where indicated) from the pre-reform steady state.

Aggregate Outcomes: Figure 8 illustrates the effects of the reform on aggregate variables. For households, the labor market effects induced by the reform lead to two opposing forces regarding precautionary savings: On the one hand, an improved labor market situation (see Figure 7) reduces the incentive to save because the likelihood of transitioning from unemployment to employment increases. On the other hand, higher income losses in the event of unemployment increase the incentive to accumulate precautionary savings. Quantitatively, the second effect dominates and aggregate asset holdings in the domestic economy rise markedly. Because the supply of domestic assets is limited by the stock of government bonds, firm profits, and domestic capital, households channel part of their additional savings into foreign bonds. As a result, the reforming country accumulates a higher net foreign asset position and the current account balance rises.

Lower wage income and increased precautionary savings create a consumption-savings trade-off, causing households to temporarily reduce their consumption. However, due to improved labor market conditions and rising wages in the medium run, the initial dip in aggregate consumption is followed by a continuous rise, stabilizing at approximately 1.5% above the pre-reform

steady state. Output and investment increase due to higher employment and an expanding capital stock. The initial response of investment plays a crucial role in moderating the impact on the current account. The rise in investment absorbs some of the surplus in savings that would otherwise have been directed toward foreign assets, thereby dampening the increase in the current account.⁴⁴

In the short run, domestic demand falls, leading to a decline in imports and contributing to the increase in net exports. Relative price adjustments reinforce expenditure switching toward domestic goods. As employment and investment expand, import demand rises and eventually exceeds its initial level, while exports increase initially and then gradually fade in the medium to long run. As a result, net exports remain above their initial level for roughly a decade before declining to a lower post-reform level.⁴⁵ Together with stronger precautionary saving in the early years, this generates a sizeable build-up of net foreign assets and temporary current account surpluses.

Government Sector: The model predicts that in response to the reform, government bonds and expenditures decline as rising labor tax revenues and lower unemployment benefits reduce the fiscal burden (see Figure 17 in Appendix C.3). This reduction in government expenditures and debt issuance leads to an increase in national savings, which in turn contributes to the rise in the current account balance. Specifically, with lower public borrowing requirements, fewer domestic savings are absorbed by government debt, allowing more saving to be allocated to foreign assets, increasing the current account balance.

This model-based prediction is consistent with the data patterns in Appendix C.3. After 2005, Germany experienced a significant reduction in its government deficit, alongside lower labor tax rates and social security contributions (see Appendix C.3, Figures 18 and 19). Furthermore, the model's prediction of increased national savings and its association with a higher current account balance is reflected in the shift in the government's contribution to Germany's current account dynamics starting in 2005, except for the period following the Great Recession (see Figure 14 in Appendix A).

Because of this interplay between the government sector and the current account, the fiscal rule plays a crucial role in shaping the overall response. In the next subsection, we perform robustness exercises regarding the fiscal rule's parameters.

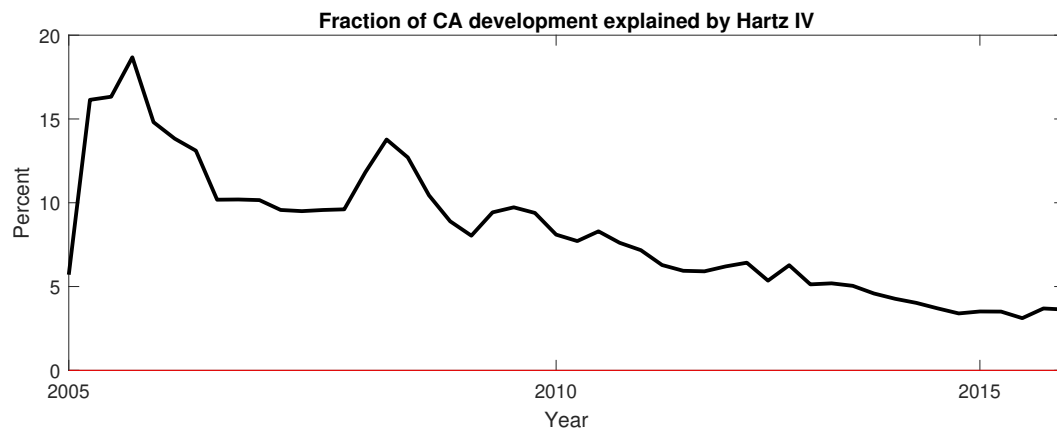
5.2 Contribution to Macroeconomic Aggregates & Robustness

This subsection assesses the impact of the Hartz IV reform on key macroeconomic aggregates and performs several robustness checks. By modeling the Hartz IV reform as the only shock, we

⁴⁴ By not introducing investment adjustment costs, our model provides a conservative estimate of the current account response. Incorporating such costs would further amplify the impact by limiting the initial surge in investment, leading to a more pronounced increase in net foreign assets and a stronger current account adjustment. We abstain from including these costs to keep the model focused on the primary mechanisms of the reform (via savings) and to maintain clarity.

⁴⁵ Consistent with steady-state balance-of-payments accounting, a permanently higher NFA position implies higher net foreign income, which must be offset by lower net exports in the long run. The law of motion $NFA_{t+1} = (1 + r - \delta)NFA_t + NX_t$ implies $\bar{N} \bar{X} = -(r - \delta)NFA$ in steady state, so higher long-run NFA requires lower long-run net exports.

Figure 9: Contribution of the Hartz IV Reform to Germany's Current Account Position (share of observed CA/GDP, in percent)



Notes: Quarterly effects on the CA/Output ratio based on our model simulations as a share of Germany's quarterly CA/GDP developments from 2005 to 2019. Data Sources: Deutsche Bundesbank and Destatis.

Table 5: Reform Contribution to Empirical Time Series over 5 and 10 years with Robustness Checks

	Baseline		Anticipation		$\rho^{\tau} = 0.9$		$\chi^b = 0.03$	
	5 years	10 years	5 years	10 years	5 years	10 years	5 years	10 years
Unemployment	41.52	21.63	40.86	21.50	40.48	20.73	41.68	21.55
Current Account/GDP	11.65	9.18	8.98	7.72	10.02	7.69	11.12	8.52
Net Foreign Assets/GDP	44.19	57.06	30.01	44.16	38.32	48.74	42.40	54.08
Real Exchange Rate	22.28	12.56	19.02	11.00	19.84	11.18	21.51	12.07

Notes: The "Baseline" column represents our standard model calibration. Unemployment effects represent the share of the unemployment rate explained by the model five (ten) years after the reform. The "Anticipation" scenario assumes agents anticipated the reform four periods in advance, with contributions including this anticipation period. In the $\rho^{\tau} = 0.9$ scenario, the fiscal rule's persistence parameter is set to 0.9. In the $\chi^b = 0.03$ scenario, the weight on government debt in the fiscal rule is 0.03.

can isolate and analyze its effects on these variables as they transition to the new steady state.⁴⁶ Figure 9 illustrates the contribution of the Hartz IV reform to Germany's current account position, expressed as a percentage of GDP, from 2005 to 2019. The graph shows that the reform had a significant initial impact, peaking at around 19% shortly after implementation and contributing to approximately 12% in the first five years after the reform. Over time, the contribution gradually decreases, fluctuating between 5% and 10%, before stabilizing at a lower level towards the end of the period.

Table 5 presents the reform's contributions to selected macroeconomic indicators over five and ten years. In the baseline (column one), the reform explains more than 40% of the unemployment decline by 2010 (five years post-reform), about 12% of the rise in Germany's current account balance, 44% of the net foreign asset increase, and 22% of real effective exchange rate changes.

Additionally, Table 5 compares the baseline scenario to several robustness checks: (i) allowing

⁴⁶ This approach differs from standard DSGE variance decomposition, which accounts for all dynamics of the observed variables using multiple shocks.

for anticipatory effects, (ii) adjusting the fiscal rule persistence to $\rho^\tau = 0.9$, and (iii) modifying the weight on government debt to $\chi^b = 0.03$. To isolate the effects of parameter changes and assess their sensitivity under the same initial conditions, we keep the shock size constant across scenarios.⁴⁷

The "Hartz IV Law" was passed on December 24, 2003, and enacted on January 1, 2005, suggesting that German households could have anticipated the reform about one year in advance. In this anticipation scenario, agents adjust their savings behavior as soon as they become aware of the upcoming reform. This early adjustment reduces the estimated impact of the reform on the current account.

Lowering the persistence parameter in the fiscal rule ($\rho^\tau = 0.9$) reduces the short-run contributions to current account dynamics. However, empirically, the adjustment of social security contributions was observed to take place two years post-reform (see Appendix C.3, Figure 19), making a higher persistence parameter more consistent with observed behavior. Finally, setting $\chi^b = 0.03$ yields results similar to the baseline, indicating that a higher elasticity with respect to debt deviations does not significantly alter the overall dynamics of the model.

Overall, the robustness checks imply that the baseline delivers the largest current account contribution for the given shock size. Anticipation and lower fiscal-rule persistence reduce the reform's contribution, while a higher debt elasticity only modestly attenuates it. Hence, within this class of specifications, our baseline contribution can be viewed as an upper bound.

Further Considerations Another concern might be that the Hartz IV reform was part of a broader reform package, including Hartz III, which aimed to improve the federal employment agency (see Appendix A for details). However, simulating the Hartz III reform, represented by an increase in matching efficiency, improves labor market outcomes without a corresponding rise in precautionary savings (see Appendix D.3 for the results). Instead, the expansionary effects on consumption and output lead to lower aggregate savings and a decline in the net foreign asset position, indicating that while Hartz III effectively reduced unemployment, it dampened the increase in net foreign assets.

6 The Role of Heterogeneity

This section assesses how well the Heterogeneous Agent (HA) model matches key empirical macroeconomic facts following the Hartz IV reforms, compared to a complete insurance model.⁴⁸ Consumer heterogeneity plays a critical role in analyzing labor market reforms for two main reasons: i) the precautionary savings motive that arises under incomplete insurance and ii) differences in agents' asset holdings, which lead to varying marginal propensities to consume (MPCs). These differences have aggregate implications, particularly when a significant share of consumers exhibit high MPCs, as noted by [Krueger et al. \(2016b\)](#). We specifically examine how

⁴⁷ As a result, we do not necessarily match the empirical target of a one percentage point increase in the aggregate private savings rate, as we do in our baseline using a regression on a simulated panel.

⁴⁸ In our analysis, we use the terms "complete insurance" and "representative agent framework" interchangeably because, in our setting, the representative agent is assumed to be fully insured against idiosyncratic shocks.

Table 6: Comparison of the Complete and Incomplete Insurance Models 5 and 10 Years Post-Reform

	Incomplete Insurance		Complete Insurance	
	5 years	10 years	5 years	10 years
Unemployment	41.52	21.63	23.35	14.90
Current Account/GDP	11.65	9.18	-4.09	-1.68
Net Foreign Assets/GDP	44.19	57.06	-22.77	-18.46
Real Exchange Rate	22.28	12.56	1.02	-0.33

Notes: Effects are computed as the average fraction of annual contributions to observed dynamics in the model relative to the data. Unemployment effects represent the share of the unemployment rate explained by the model five (ten) years after the reform.

both models account for dynamics in labor markets, savings, and external balances, and we also explore the redistributive effects arising from these reforms.

6.1 Comparison to the Complete-Insurance Framework

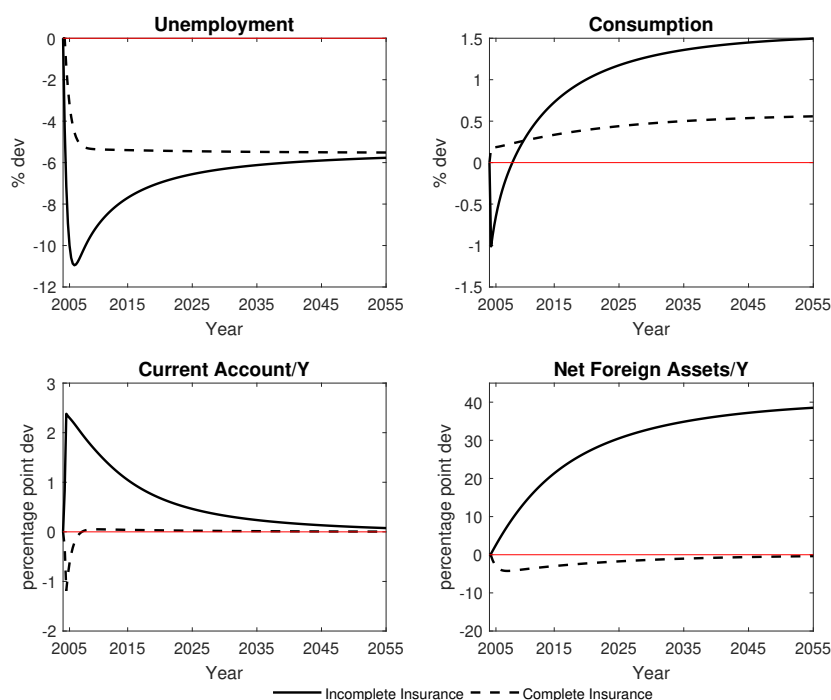
We simulate the reform using a version of our model with a representative agent framework with full insurance against idiosyncratic unemployment risk. In this framework, consumption, and savings decisions are independent of employment status, following [Andolfatto \(1996\)](#) and [Merz \(1995\)](#). Appendix C.4 details the household's problem in this case. As with the baseline heterogeneous agent model, we simulate the complete insurance model non-linearly under perfect foresight.⁴⁹

Table 6 compares the performance of the heterogeneous agent (HA) and complete insurance (representative agent, RA) models in explaining (international) macroeconomic variables five and ten years after the reform. The HA model better captures post-reform dynamics, particularly for the current account, net foreign assets (NFA/GDP), and the real exchange rate (REER). These results suggest the HA model explains a larger fraction of observed adjustments in household savings and external balances. By contrast, the RA model generates *qualitatively* different outcomes for the current account and net foreign assets and contributes *quantitatively* less to observed external-balance and exchange-rate dynamics.

Figure 10 contrasts the simulation results from our heterogeneous agent model (solid line) with the complete insurance case (dashed line), showing qualitatively different responses for the current account and the net foreign asset position. The main difference can be explained by the absence of a precautionary savings motive in the complete insurance model. In the HA setting, consumption declines in the short run as agents start accumulating savings, reflecting a consumption-savings trade-off. In the complete insurance model, consumption increases immediately and savings decrease slightly following the reform due to improved labor market conditions (higher job-finding rate and employment). As a result, domestic demand effects dominate, leading to a fall in the current account balance and the net foreign asset position. Consequently, when households are perfectly insured against unemployment-related consumption

⁴⁹ To ensure stationarity in the complete-insurance framework, we adopt a debt-elastic interest rate following [Schmitt-Grohe and Uribe \(2003\)](#); see Appendix C.4 for details.

Figure 10: Comparison to the Complete Insurance Framework



Notes: Model responses in the baseline heterogeneous agent (HA) model (solid line) and the complete insurance (RA) model (dashed line). The shock size is calibrated to achieve the same steady state effect on unemployment as in the HA model.

risk, a reduction in the replacement rate and a cut in entitlement duration have the opposite implication for net foreign assets.⁵⁰

6.2 Redistributive Effects

Table 7 illustrates the effects of the Hartz IV labor market reform on wealth and consumption inequality in Germany. The Gini coefficient for wealth declines across all employment and skill groups, reflecting a reduction in wealth inequality. This is primarily driven by an increase in savings and asset accumulation among low-skilled workers, who move further from the borrowing constraint due to the reform. This pattern aligns with the observed decline in Germany's wealth Gini starting in 2008, as documented by [Albers, Bartels, and Schularick \(2022\)](#).

Consumption inequality, measured by the coefficient of variation, shows heterogeneous responses to the reform. For high-skilled employed and short-term unemployed (STU) workers, consumption variability increases slightly, while it decreases for low-skilled workers and long-term unemployed individuals. The strong reduction in consumption variability for long-term unemployed workers is driven by the compression in consumption levels as all long-term unemployed workers are directly affected by the reform and face a decline in their consumption levels. In contrast, for high-skilled workers, improved labor market conditions and higher savings lead to greater dispersion in consumption levels, and thereby higher consumption variability.

⁵⁰ The size of the shock is calibrated to produce the same steady state effect on unemployment as in the incomplete insurance baseline. This approach is necessary because the savings rate decreases in the short run due to the absence of a precautionary savings motive, making it unsuitable as a dynamic target.

Table 7: Change in Inequality by Employment Status and Skill Level

	Pre-Reform	Post-Reform	Percentage Change
Gini Coefficient			
Low-skilled Employed	0.81	0.75	-7.45
High-skilled Employed	0.41	0.40	-1.87
Low-skilled STU	0.78	0.73	-6.40
High-skilled STU	0.41	0.40	-1.93
Long-term Unemployed	0.82	0.80	-3.03
Coefficient of Variation for Consumption			
Low-skilled Employed	2.85	2.73	-4.50
High-skilled Employed	1.02	1.04	1.65
Low-skilled STU	2.55	2.27	-10.87
High-skilled STU	1.05	1.07	1.44
Long-term Unemployed	9.73	4.57	-53.06

Notes: The table displays within-group Gini coefficients and coefficients of variation for consumption in the pre-reform steady state, the post-reform steady state, and the percentage change between steady states by employment and skill group. "STU" refers to short-term unemployed workers.

Our findings on the reform's redistributive effects and the associated increase in the country's net foreign asset position align with [de Ferra et al. \(2021\)](#), who show that, in advanced economies, capital tends to flow from more equal to less equal countries. In our setting, the post-reform decline in inequality is therefore accompanied by net capital outflows.

7 Welfare Analysis

This section examines the welfare implications of reduced unemployment benefits across several dimensions. We begin by analyzing the welfare effects during the transition to the post-reform steady state, followed by a comparison of steady state welfare outcomes. The analysis also contrasts the results with those from a model incorporating complete insurance and evaluates the differences within a closed-economy framework. Lastly, we decompose the welfare effects into partial equilibrium, matching, and price effects, following the approach of [Mukoyama \(2013\)](#). We quantify the relative welfare consequences of the German unemployment benefit reform for consumers in all skill groups and employment states along the wealth distribution. For this purpose, we calculate welfare effects in terms of consumption equivalents (CE) after the reform. We calculate the CE for employment state i as

$$CE^i(a; Z) = \left(\frac{V^{i,R}(a; Z)}{V^{i,0}(a; Z)} \right)^{\frac{1}{1-\sigma}} - 1,$$

where $V^{i,R}(a; Z)$ denotes the value function after implementing the reform and $V^{i,0}(a; Z)$ is the initial value function in steady state.⁵¹

⁵¹ To calculate the effects along the transition path, we follow [Bayer et al. \(2019\)](#) and compute the value functions $V^{i,R}(a; Z)$ by iterating backward given the equilibrium paths of the wage, prices, labor income tax, and the job-finding rate. We calculate consumption equivalents based on the value function $V^{i,R}(a; Z)$ one period after the shock implementation.

7.1 Transitional vs. Steady State Effects

Table 8 illustrates the welfare effects of the labor market reform in terms of consumption equivalents by wealth quintiles, employment state, and skill group.

The reform generates welfare losses for most low-skilled workers, particularly those with lower asset holdings, during the transition. These workers rely heavily on labor income, have high marginal propensities to consume, and face declining wages, making them more vulnerable. In contrast, asset-rich individuals benefit from their ability to smooth consumption, mitigating the impact of the reform. The largest welfare losses are experienced by low-skilled workers, as well as short-term and long-term unemployed individuals.

In steady state, these welfare losses become even more pronounced for low-skilled and long-term unemployed workers. Their income declines, and they are directly affected by the reduction in consumption insurance (see the discussion in the next subsection). Additionally, wages for low-skilled workers decrease further, while wages for high-skilled workers increase slightly in the post-reform steady state.

While the overall welfare effects of the reform are modest, the distribution of gains and losses varies significantly across the population. In contrast, under complete insurance, there is no consumption-savings trade-off, as consumption immediately increases due to positive labor market effects (as depicted in Figure 10). Consequently, welfare effects are uniformly positive in the complete-insurance version (0.36 during the transition and 0.59 in steady state comparisons; not reported in Table 8). This again highlights that abstracting from precautionary savings leads to *qualitatively* different effects.

Comparison to the Closed Economy Next, we compare the welfare effects of the reform in a small open economy to those in a closed economy setting, which lacks access to international bonds and features an endogenous interest rate.⁵² A key distinction between the two settings is that aggregate savings rise more sharply in the open economy because consumers have access to international bonds, while domestic asset supply adjusts more slowly (see Figure 21 in Appendix D.4). In the closed economy, this excess demand for assets results in a temporary increase in the interest rate. Over the long run, the closed economy experiences an increase in aggregate asset supply and capital stock, which ultimately reduces the interest rate in the new steady state. This more muted response in savings also leads to a quicker recovery in aggregate consumption. These differences across frameworks are reflected in the welfare effects. During the transition, the temporary rise in the interest rate and stronger labor market responses (see Figure 21) lead to slightly greater welfare gains and less severe welfare losses in the closed economy. In steady state, the average welfare effect remains negative in the small open economy but turns positive in the closed economy, driven by a stronger increase in the job-finding rate in the latter.

⁵² Appendix D.4 illustrates the corresponding dynamic effects and welfare decompositions in steady state for the small open vs. closed economy.

Table 8: Welfare Effects by Wealth Quintiles in the Small open vs. Closed Economy

	Wealth Quintiles						AVG	Economy AVG
	1	2	3	4	5			
Transitional Effects								
Small open economy								
Low-skilled Employed	-0.25	-0.20	-0.18	-0.03	0.37	-0.06	-0.03	
High-skilled Employed	0.45	0.48	0.47	0.44	0.39	0.45		
ST Unemployed, low skilled	-1.00	-0.86	-0.73	-0.35	0.35	-0.52		
ST Unemployed, high skilled	0.21	0.55	0.64	0.67	0.65	0.55		
LT Unemployed	-3.92	-3.78	-3.19	-1.32	0.02	-2.45		
Closed Economy								
Low-skilled Employed	0.06	0.11	0.13	0.28	0.51	0.22	0.15	
High-skilled Employed	0.58	0.52	0.42	0.30	0.11	0.38		
ST Unemployed, low skilled	-0.70	-0.56	-0.42	-0.04	0.51	-0.24		
ST Unemployed, high skilled	0.50	0.77	0.77	0.69	0.51	0.65		
LT Unemployed	-3.64	-3.50	-2.90	-1.02	0.13	-2.20		
Steady State Effects								
Small open economy								
Low-skilled Employed	-0.25	-0.22	-0.20	-0.07	0.30	-0.08	-0.08	
High-skilled Employed	0.51	0.52	0.51	0.48	0.42	0.49		
ST Unemployed, low skilled	-1.39	-1.27	-1.15	-0.74	0.12	-0.88		
ST Unemployed, high skilled	-0.11	0.26	0.37	0.42	0.44	0.28		
LT Unemployed	-4.33	-4.21	-3.68	-1.79	-0.24	-2.86		
Closed Economy								
Low-skilled Employed	0.09	0.12	0.14	0.24	0.37	0.19	0.10	
High-skilled Employed	0.60	0.53	0.43	0.32	0.14	0.40		
ST Unemployed, low skilled	-0.95	-0.83	-0.72	-0.35	0.23	-0.52		
ST Unemployed, high skilled	0.19	0.44	0.45	0.40	0.27	0.35		
LT Unemployed	-3.85	-3.74	-3.21	-1.40	-0.16	-2.48		

Notes: Effects are measured in terms of consumption equivalents. Transition effects refer to consumption equivalents in the first period after the shock implementation. Steady state effects compare consumption equivalents between pre-reform and post-reform equilibria. The last column reports the economy-wide population-average welfare effect.

7.2 Decomposing the Welfare Effects

To gain a clearer understanding of the welfare impacts, we decompose the steady state welfare effects into three key components: the partial equilibrium effect, the matching effect, and the price effect.⁵³ Figures 11 (for employed consumers) and 12 (for unemployed consumers) illustrate the results.

The *partial equilibrium effect*, represented by the blue dashed line in Figures 11 (for employed consumers) and 12 (for unemployed consumers), isolates the impact of changes in unemployment benefits and taxes.⁵⁴ When the reform is implemented, both the replacement rate for long-term unemployment benefits and the lump-sum tax are reduced. The overall effect is mostly negative for unemployed consumers due to reduced consumption insurance, which particularly impacts asset-poor, long-term unemployed individuals. However, for employed consumers—especially those who are high-skilled and asset-rich—the reduced tax burden acts as

⁵³ This decomposition follows the methodology of Mukoyama (2013), while potential non-linear interactions are not addressed here.

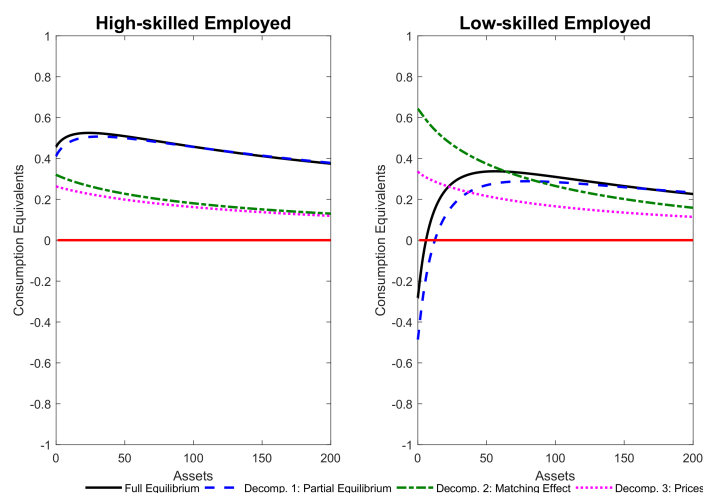
⁵⁴ In this analysis, the transition probability between short- and long-term unemployment remains constant.

an implicit transfer, mitigating the negative effects and resulting in positive welfare gains for them.

The *matching effect*, depicted by the green dotted line, arises from changes in the job-finding rate. For this analysis, we compute the welfare effects assuming that consumers experience only the transition path of the job-finding rate as modeled in our baseline simulation, while other variables remain fixed at their pre-reform steady state levels. As the reform increases the job-finding rate, welfare improves for all consumers, with the effect being particularly strong for those who are currently unemployed. Asset-poor consumers, who are more dependent on labor income, benefit the most from the matching effect.

The *price effect*, shown by the magenta line, reflects general-equilibrium adjustments in wages and relative goods prices. In the new steady state, the reform slightly raises the steady state average wage and lowers the relative price of foreign goods, implying higher real purchasing power.⁵⁵ This gain is concentrated among asset-poor households, for whom wage income constitutes a larger share of resources and consumption expenditures represent a larger share of their budget.

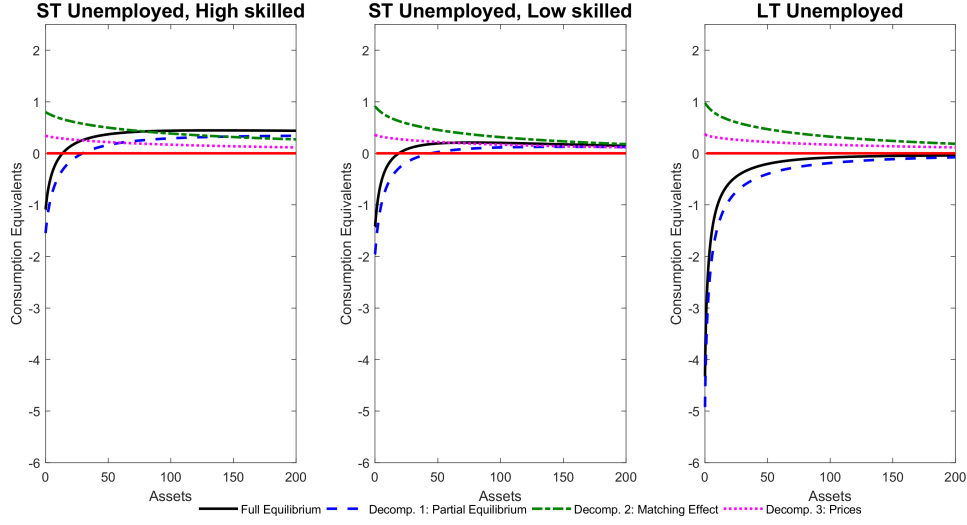
Figure 11: Welfare Decomposition for Employed Consumers: Stationary Effects



Notes: Responses to the labor market reform in the small open economy (solid line), the partial equilibrium model (only changes in benefits, illustrated by the blue dashed line), the matching effect (effects via changes in the job-finding rate, illustrated by the green dotted line), and the price effect (effects via wage and goods price reactions, illustrated by the magenta dotted line).

⁵⁵ See Table 12 in Appendix D.2.

Figure 12: Welfare Decomposition for Unemployed Consumers: Stationary Effects



Notes: Responses to the labor market reform in the small open economy (solid line), the partial equilibrium model (only changes in benefits and the lump-sum tax, illustrated by the blue dashed line), the matching effect (effects via changes in the job-finding rate, illustrated by the green dotted line), and the price effect (effects via wage and goods price reactions, illustrated by the magenta dotted line).

8 Concluding Remarks

This paper investigates the role of precautionary savings and consumer heterogeneity in evaluating the open-economy effects of labor market reforms. We establish a link between the reduction in the generosity of unemployment benefits and the reforming country's net foreign asset position via a precautionary savings channel. While the effects of labor market reforms are well explored in closed-economy incomplete-market models, their consequences in an open-economy setting have gained little attention in the academic literature. We fill this gap by building a small open-economy heterogeneous agent model with labor market frictions.

We apply our framework to simulate the far-reaching unemployment benefit reform in Germany (Hartz IV) and find that the reduction in the generosity of the unemployment insurance scheme increases precautionary savings significantly. Because domestic asset supply is limited, part of the additional saving is invested abroad, increasing the net foreign asset position and the current account balance. Our empirical analysis further supports this finding, showing that the most affected groups significantly increased their savings rates after the reforms. We find that, in our baseline, the reform accounts for up to about 12% of the observed change in Germany's current account over the first five years. Comparing our baseline incomplete insurance model to a setting with a representative agent and complete insurance, we find that accounting for the precautionary savings channel is qualitatively and quantitatively important to capture (international) macroeconomic dynamics in the aftermath of the reform. Thus, we document that allowing for precautionary savings is crucial when evaluating labor market reforms in open economies—an insight that extends beyond the specific case of Germany examined in this study.

We analyze how the structural reform affects consumers across the wealth distribution and find that welfare gains and losses are unevenly distributed. During the transition, low-skilled, asset-

poor workers face significant welfare losses due to declining wages and foregone consumption. Asset-rich individuals, however, benefit from increased job-finding rates and their ability to smooth consumption.

Several open questions remain for future research. Possible extensions include incorporating a portfolio-choice problem and an incentive to hold foreign assets due to higher returns or a lower correlation with the domestic business cycle. Beyond precautionary saving, other margins of self-insurance, such as reduced quits or shifts toward safer jobs, may also respond to reforms. We leave these topics for future work.

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A Background

This section briefly outlines the background of Germany's current account and net foreign asset position. We also summarize the main points of the cluster of labor market reforms, known as the Hartz reforms, implemented in Germany between 2003 and 2006.

A.1 The German Current Account and Net Foreign Asset Position

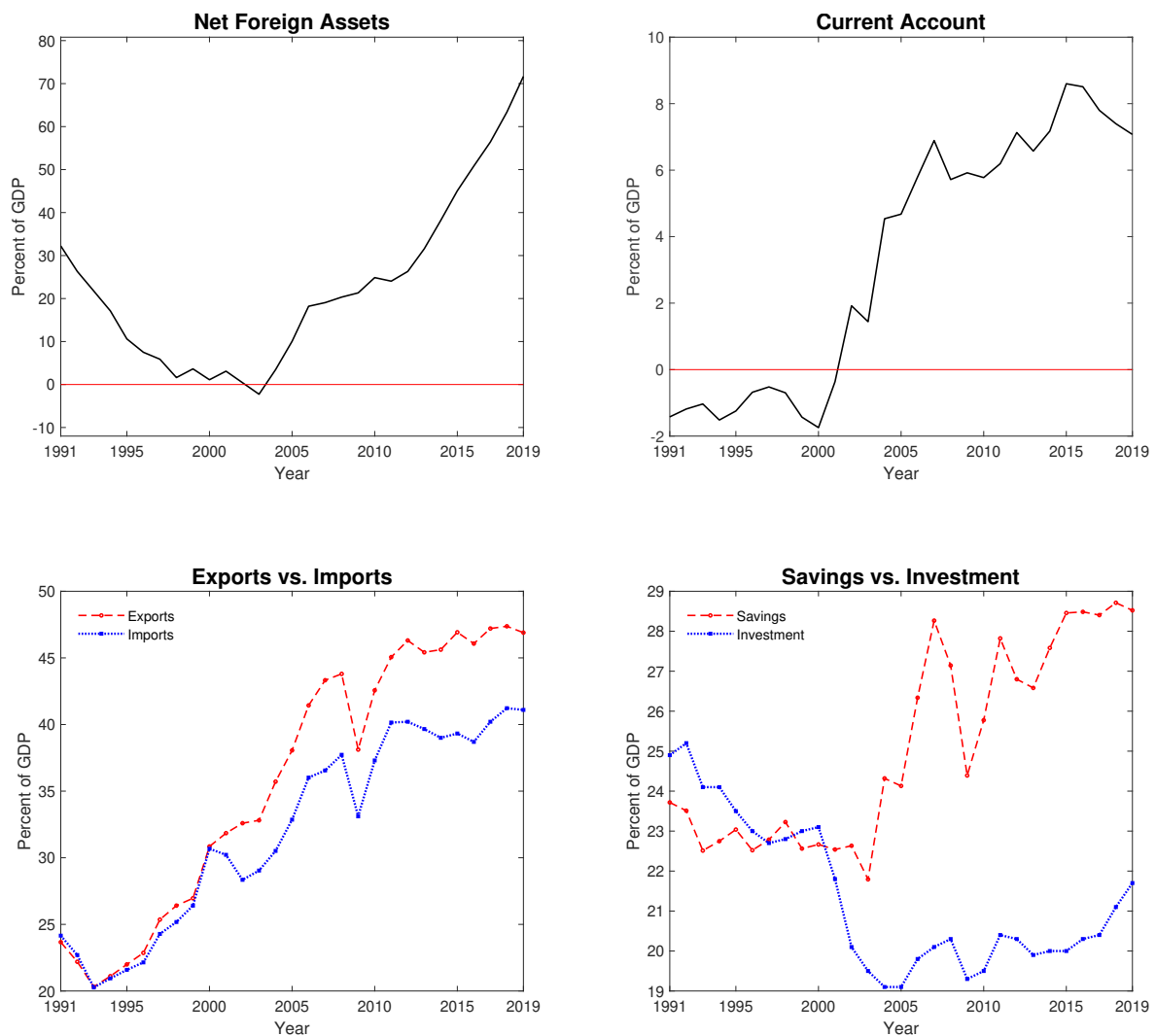
Figure 13 shows the German net foreign asset position (*NFA*), current account (*CA*), as well as exports (*EX*) and imports (*IM*), and savings and investment from 1991 onwards. Between 1991 and the early 2000s, a decade that was characterized by high unemployment rates and low GDP growth, Germany has repeatedly been called 'the sick man of Europe' (see, for example, [The Economist, 2017a](#)). Even though Germany had current account surpluses before the German Reunification of around 4 percent of GDP, between 1991 and 2000, there were no surpluses worth mentioning. However, starting in 2001, the German economy experienced a reversal: From the early 2000s onward, Germany's external balance shifted markedly: exports began to exceed imports on a sustained basis. At the same time, national saving increased relative to investment. By the accounting identity $CA = EX - IM = S - I$, this corresponds to persistent current account surpluses and a rising net foreign asset position. Germany's net foreign assets reached about 51% of GDP in 2016, making it a sizable net external creditor. These developments have attracted substantial international criticism (see, for example, Eichengreen's comment in [The Guardian, 2017](#), and [The Economist, 2017b](#)).

Figure 14 illustrates the components of the German current account as a percentage of GDP from 1991 to 2019. It breaks down the current account into contributions from financial corporations (in blue), the government (in red), private households (in yellow), and non-financial corporations (in purple). Over the period, the figure shows a significant shift in the contributions of these sectors. Notably, after the early 2000s, private households and non-financial corporations increasingly contributed to Germany's current account surplus, while the government and financial corporations exhibited smaller or negative contributions.

A.2 The Hartz Reforms

Germany's weak economic performance around the 2000s motivated a comprehensive reform package. The centerpiece of the reform agenda was a set of extensive labor market reforms, commonly known as the "Hartz reforms" (for a detailed description of the Hartz reforms, see [Jacobi and Kluve, 2006](#)). Their objectives were to improve job matching efficiency and incentives to take up employment (Hartz I), promote the transition to self-employment and introduce more flexible arrangements for minor employment relationships (Hartz II), and further support the matching process between firms and workers through a reorganization of the Federal Labor Agency (Hartz III). Table 9 provides an overview of the key changes of the corresponding reforms. In 2005, the far-reaching and most discussed Hartz IV reform was implemented. Before Hartz IV, short-term unemployed workers were entitled to unemployment benefits of up to 67 percent of their net previous salary ("*Arbeitslosengeld*"). The duration of short-term unemployment bene-

Figure 13: Net Foreign Assets, Current Account, Exports and Imports, and National Savings and Investment for Germany, 1991-2019



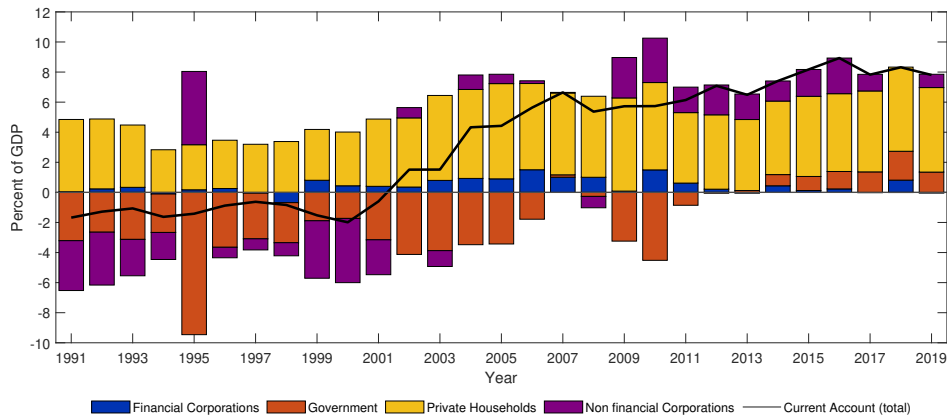
Notes: Values are in percent of GDP. Savings refer to gross national savings and are defined as disposable income minus consumption and net transfers. Exports and Imports refer to both goods and services. Data sources: Destatis (2021) and Bundesbank (2021).

fits depended on age and contribution history, with a maximum of 32 months before the reform. Unemployed workers were then considered long-term unemployed and received a less generous unemployment benefit ("*Arbeitslosenhilfe*") amounting to 57 percent of their previously earned net wage.⁵⁶ Persons who were not eligible for unemployment benefits received means-tested social assistance ("*Sozialhilfe*"; in 2004, the standard rate for a single household was around 300 Euros, not including one-time benefits).

The Hartz IV reform had two components: First, social assistance and long-term unemployment benefits were merged into "*Arbeitslosengeld II*" (*ALG II*). Hence, from 2005 onwards, long-term unemployment benefits were independent of previous earnings. Second, the maximum entitlement duration of short-term unemployment benefit receipt was reduced. The severity of the

⁵⁶ This replacement rate refers to unemployed workers with children.

Figure 14: Components of the German Current Account



entitlement cut differed by age. Before the reform, the maximum duration was 12 months for workers younger than 45 years and up to 32 months for workers older than 56 years. In 2006, when the reform took effect, the maximum duration of entitlement to unemployment benefits was 12 months for workers under 55 and 18 months for older workers. In 2008, the maximum duration for older workers was again extended to 24 months.

Table 9: Overview of the Hartz Reforms

Hartz Reform	Implementation Year	Key Changes
Hartz I	2003	Facilitated temporary employment and introduced training vouchers.
Hartz II	2003	Introduced new marginal employment types (e.g., Mini-jobs and Midi-jobs) and start-up subsidies ("Ich-AG").
Hartz III	2004	Restructured the Federal Employment Agency.
Hartz IV	2005 & 2006	Merged unemployment and social assistance into a lower, means-tested benefit (ALG II) after a shortened duration of higher unemployment benefits (ALG I), with sanctioning measures. The reduction in entitlement duration was implemented in 2006.

Asset Limits Long-term unemployment benefits in Germany are subject to means testing, with asset limits that vary by age. The maximum limit is approximately 10,000 Euros per adult and 3,100 Euros per child, with an additional cap of 48,750 Euros for retirement assets. According to the Panel of Household Finance provided by the Deutsche Bundesbank (wave 1 for 2010/2011), the second quintile (20-40% of the net wealth distribution) owns a median net wealth of 11,660 Euros per household. Therefore, these asset limits are not binding for a significant fraction of German households.

Furthermore, data from the IAB Panel Labour Market and Social Security (PASS) — a panel survey of approximately 10,000 households, starting in 2007, with a particular focus on long-term unemployed individuals and ALG II recipients — shows that only 12 percent of households receiving short-term unemployment benefits (ALG I) have savings exceeding 10,000 Euros. Addi-

tionally, around 10 percent of applicants for long-term unemployment benefits (ALG II) had to deplete their savings to meet eligibility requirements, while 60 percent qualified for assistance after three months. This suggests that, for most households, the eligibility criteria do not create strong distortions in savings incentives.

B Further Empirical Evidence

B.1 Micro Evidence on Private Savings of Full-time Employed

We analyze the private savings rate of full-time employed household heads using data from the German Socioeconomic Panel (GSOEP) spanning the years 1995 to 2014. The GSOEP is a comprehensive longitudinal survey that includes approximately 11,000 private households, with data collection starting in 1984 in West Germany and including East Germany since 1990. Further details on the GSOEP dataset can be found in [Goebel, Grabka, Liebig, Kroh, Richter, Schröder, and Schupp \(2018\)](#).

To construct the private savings rate presented in Figure 2, we adhere to the methodology outlined by [Stein \(2009\)](#). Initially, we exclude all observations that do not correspond to household heads. We then calculate the annual total of weighted monthly savings (variable *hlc0120*) and divide this by the annual total of weighted monthly household net income (variable *hlc0005_h*) to derive the private savings rate. To ensure that our savings rates are representative of the broader population, we apply cross-sectional household weights (variable *hhref*).

B.1.1 Saving Rates by Educational Groups

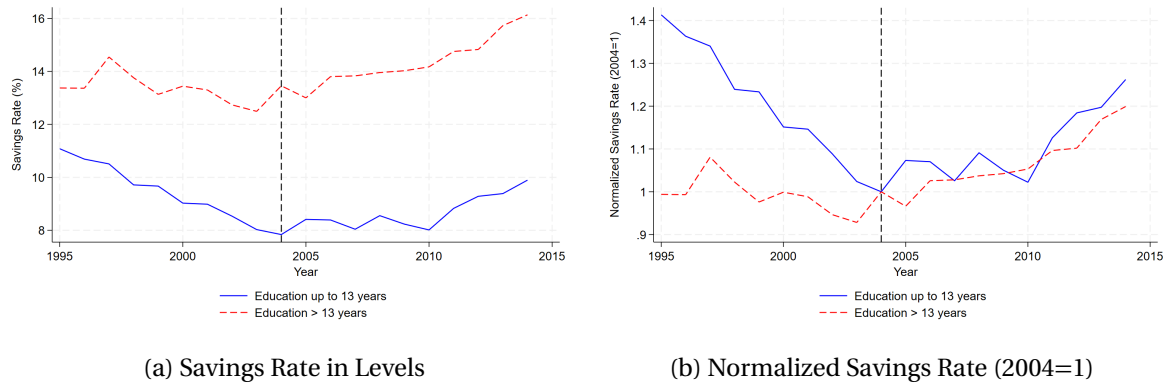
Figure 15 illustrates the evolution of private savings rates by education group from 1995 to 2014. Panel (a) presents savings rates in levels, showing that individuals with more than 13 years of education consistently saved at a higher rate than those with fewer years of education. Panel (b) normalizes savings rates to 2004 (2004=1), the year before the reform was implemented (indicated by the vertical dashed line), highlighting differences in savings behavior after the reform. The savings rate of low-educated workers initially increased but declined during the Great Recession, indicating greater sensitivity to business cycle fluctuations. In contrast, the savings rate of highly educated workers rose more gradually before accelerating after 2010.

B.2 Cross-Country Evidence: Current Account Movements in European Countries

Figure 16 shows annual current account-to-GDP ratios for Germany, Austria, Italy, Spain, Portugal, and France from 1995 to 2014. Germany's current account balance (blue) begins to rise in the early 2000s, preceding the major labor market reforms. This timing suggests that factors such as the introduction of the Euro and the convergence of interest rates within the Eurozone (see, e.g., [Berger and Nitsch, 2010](#)) likely contributed to the early dynamics. After the reform period, the current account-to-GDP ratio continues to increase.

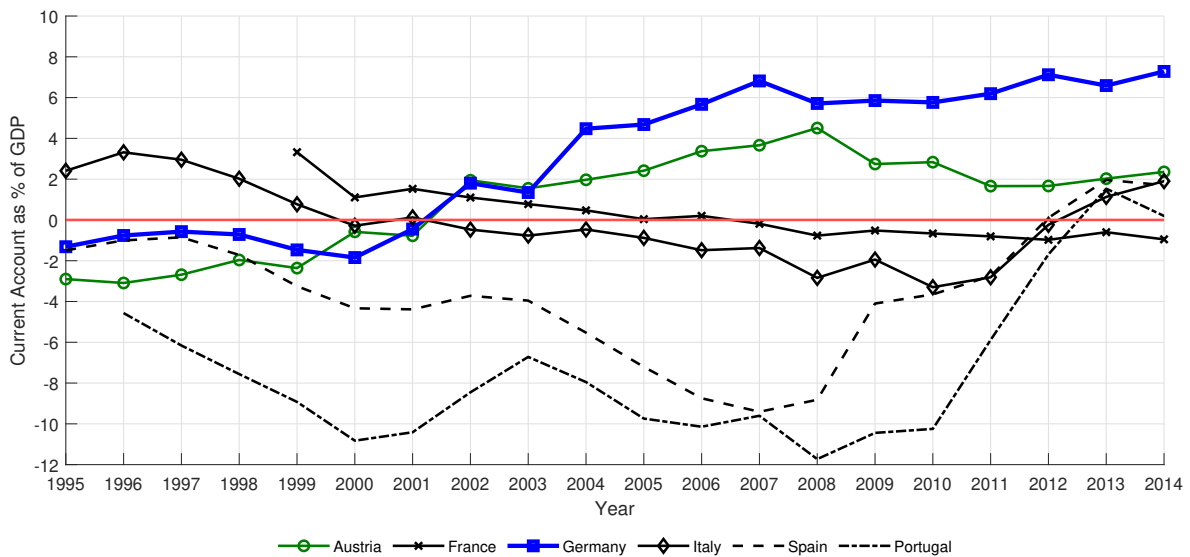
Austria (dark green) shares several institutional similarities with Germany, but did not implement comparable labor market reforms. Although Austria's current account balance rose around

Figure 15: Private Savings Rates by Educational Group, 1995-2014



Notes: The left panel (a) shows the private savings rate by education group in levels from 1995 to 2014. The right panel (b) presents the normalized savings rate, with values indexed to 2004 (2004=1). The solid blue line represents individuals with up to 13 years of education, while the dashed red line represents those with more than 13 years of education. The vertical black dashed line marks the year 2004 (right before the reform was implemented).

Figure 16: Current Account-to-GDP Ratio for Selected European Countries, 1995-2014



Notes: The figure displays the current account-to-GDP ratio for Germany, Austria, France, Italy, Portugal, and Spain from 1995 to 2014. The blue (green) line represents Germany (Austria). Data Source: OECD (2023)

the introduction of the Euro in 2000, the increase was smaller and less persistent than in Germany thereafter.

Other major Eurozone economies exhibit different patterns. Spain, Italy, and Portugal record persistent deficits over much of the sample, while France's balance remains close to zero.

Overall, Germany's current account dynamics over this period stand out in both magnitude and persistence. As [Jaumotte and Sodsriwiboon \(2010\)](#) notes, several Southern European economies ran deficits after the introduction of the Euro, a pattern linked in part to declining private saving; this contrasts with Germany, where private saving rose over the same period.

B.3 Time Series Evidence of the Hartz IV Reform on the Current Account

In this subsection, we analyze the impact of the German Hartz IV reform on the current account-to-GDP ratio using an AR(1) time series model. Table 10 presents the results of this analysis, with models incorporating various macroeconomic variables (beyond one lag of the dependent variable) to control for factors that could influence the current account. These include the interest rate, the CPI-based real effective exchange rate (REER), the export price index, export growth, the terms of trade, and lagged government debt.

The Hartz IV reform is represented by a dummy variable that takes the value 1 from 2005 onward. The reform shows a statistically significant positive effect on the current account-to-GDP ratio. This is consistent when varying the set of controls. In our baseline specification (Model 1), the reform is associated with a 1.75 percentage point increase in the current account-to-GDP ratio, a finding that remains robust across alternative model specifications.

Finally, Model 4 examines rolling 5-year windows. The largest and most precisely estimated effect appears in the first five years after implementation (2005–2009), when the reform dummy is associated with a 2.81 percentage point increase in CA/GDP . Taken at face value, these estimates imply a sizable rise in CA/GDP after 2005, conditional on the included controls.

Table 10: Effects of the German Labor Market Reform on the Current Account

	Model 1	Model 2	Model 3	Model 4
<i>Dependent Variable: CA/GDP</i>	Dummy: 2005-14	Dummy: 2005-14	Dummy: 2005-14	5-year windows
Interest rate	-0.455*** (0.116)	-0.215 (0.141)	-0.445** (0.189)	-0.471*** (0.139)
GDP Growth	-0.008 (0.070)	-0.039 (0.074)	0.009 (0.105)	-0.006 (0.066)
REER (CPI-based)	0.141*** (0.052)		0.235** (0.095)	0.138*** (0.051)
Export price index	0.417*** (0.088)	0.450*** (0.098)	0.515*** (0.174)	0.463*** (0.082)
Export growth	0.096*** (0.033)	0.099*** (0.033)	0.100** (0.046)	0.092*** (0.032)
Hartz IV - Dummy	1.748*** (0.353)	1.690*** (0.327)	1.865*** (0.454)	
Terms of Trade		0.210** (0.087)		
Lag (log Government Debt,1)			-2.286 (4.392)	
Dummy 2000-04				0.867 (0.990)
Dummy 2005-09				2.807** (1.219)
Dummy 2010-14				1.681 (1.093)
Lagged CA/GDP	0.889*** (0.053)	0.893*** (0.049)	0.809*** (0.099)	0.811*** (0.118)
Constant	-49.627*** (11.713)	0.632*** (0.064)	-59.038*** (16.263)	-53.869*** (11.036)
Observations	84	84	63	84
R ²	0.983	0.982	0.972	0.967
Wald chi2	1164.70	914.84	342.75	760.34
Prob > chi2	0.00	0.00	0.00	0.00
AIC	180.93	183.69	151.20	180.95
BIC	202.80	205.56	172.63	207.69

Note: "Hartz IV Dummy" is a shift dummy that takes the value 1 from 2005 onward. The quarterly sample ranges from 1994/Q1 to 2014/Q1. Due to data availability, the sample for Model 3, including Government debt, ranges from 1999/Q1 to 2014/Q4. Model 4 contains 5-year windows. *p<0.1; **p<0.05; ***p<0.01.

C Model Appendix

C.1 The Investment Fund

To keep track of aggregate asset holdings and impose standard no-arbitrage conditions in an economy with multiple traded claims, it is convenient to introduce a *notional* investment fund that consolidates the balance sheets of domestic savers. The fund simply implements the decomposition of households' one-dimensional asset position into a portfolio of traded assets consistent with market clearing and no-arbitrage.

Pricing kernel. With no aggregate uncertainty and an exogenous gross return R on a CPI-denominated risk-free claim, the pricing kernel is deterministic:

$$m_{t,t+1} = \frac{1}{R} \quad \text{for all } t, \quad \text{so } m_{0,t} = R^{-t},$$

with $R = 1 + r - \delta$, where r is the (constant) rental rate on capital and δ is the depreciation rate. Hence, $m_{0,t}$ coincides with the present-value operator implied by the world risk-free return and is common across households in this deterministic environment.

Asset decomposition and timing. The fund allocates aggregate household assets $\bar{a}_t$ across four asset classes: physical capital k_t (valued at $\frac{p_t^H}{P_t}$ in CPI units), government bonds b_t , net foreign assets NFA_t , and equity (a claim on aggregate firm profits). We normalize the aggregate supply of equity to one share, so $x_t = 1$.

The market value of equity held by domestic households at the beginning of period t is simply the predetermined price p_{t-1} . The corresponding balance-sheet identity (asset market clearing) is

$$\bar{a}_t = \frac{p_t^H}{P_t} k_t + b_t + NFA_t + p_{t-1}. \quad (\text{C.1})$$

Equity pricing. Let d_t denote aggregate dividends paid in period t . Holding one unit of equity from t to $t+1$ yields the gross payoff $d_{t+1} + p_{t+1}$. No-arbitrage with the pricing kernel implies the standard present-value condition

$$p_t = \frac{d_{t+1} + p_{t+1}}{R}.$$

No-arbitrage across asset classes. Because there is no aggregate risk, CPI-denominated one-period claims are risk-free from the aggregate perspective. Hence, such claims are priced using the same deterministic discount factor implied by the world risk-free return R . In particular,

$$1 + r_G = R, \quad 1 + r_{NFA} = R,$$

where r_G denotes the return on a government bond and r_{NFA} the return on NFA (both in CPI units), so $r_G = r_{NFA} = R - 1$. Physical capital is valued on the fund's balance sheet at replacement

cost in CPI units, $q_t^k k_t$ with $q_t^k \equiv \frac{p_t^H}{P_t}$; movements in q_t^k therefore track the capital position in CPI terms along the deterministic transition.

Interpretation. Households choose only the level of the one-dimensional asset a_t . Given $\bar{a}_t$, the investment fund's role is purely to implement a portfolio $\{\frac{p_t^H}{P_t} k_t, b_t, NFA_t, p_{t-1}\}$ satisfying (C.1) and the no-arbitrage restrictions above. Therefore, there is no risk–return trade-off across asset classes in the model.

C.2 Derivation of the Resource Constraint

We start with the aggregate budget constraint of all households:

$$C_t + \bar{a}_{t+1} = (1 - \tau^w) w_t N_t + \kappa_t^{ST,H} U_t^{ST,H} + \kappa_t^{ST,L} U_t^{ST,L} + \kappa_t^{LT} U_t^{LT} - \tau_t + (1 + r - \delta) \bar{a}_t, \quad (C.2)$$

where $w_t N_t \equiv w_t^L N_t^L + w_t^H N_t^H$ denotes the aggregate wage bill. Next, we plug in the asset market clearing condition:

$$\begin{aligned} & \overbrace{C_t + NFA_{t+1} + \frac{p_{t+1}^H}{P_{t+1}} k_{t+1} + b_{t+1} + p_t}^{\bar{a}_{t+1}} = (1 - \tau^w) w_t N_t + \kappa_t^{ST,H} U_t^{ST,H} + \kappa_t^{ST,L} U_t^{ST,L} + \kappa_t^{LT} U_t^{LT} \\ & - \tau_t + (1 + r - \delta) \overbrace{(NFA_t + \frac{p_t^H}{P_t} k_t + b_t + p_{t-1})}^{\bar{a}_t} \end{aligned} \quad (C.3)$$

Next, we substitute out the predetermined equity position p_{t-1}

$$\begin{aligned} & C_t + NFA_{t+1} + \frac{p_{t+1}^H}{P_{t+1}} k_{t+1} + b_{t+1} + p_t = (1 - \tau^w) w_t N_t + \kappa_t^{ST,H} U_t^{ST,H} + \kappa_t^{ST,L} U_t^{ST,L} + \kappa_t^{LT} U_t^{LT} \\ & - \tau_t + (1 + r - \delta) (NFA_t + \frac{p_t^H}{P_t} k_t + b_t + \overbrace{\frac{d_t + p_t}{1 + r - \delta}}^{p_{t-1}}) \end{aligned} \quad (C.4)$$

Rearrange slightly and see that p_t cancels out.

$$\begin{aligned} & C_t + NFA_{t+1} + \frac{p_{t+1}^H}{P_{t+1}} k_{t+1} + b_{t+1} = (1 - \tau^w) w_t N_t + \kappa_t^{ST,H} U_t^{ST,H} + \kappa_t^{ST,L} U_t^{ST,L} + \kappa_t^{LT} U_t^{LT} \\ & - \tau_t + (1 + r - \delta) (NFA_t + \frac{p_t^H}{P_t} k_t + b_t) + d_t \end{aligned} \quad (C.5)$$

Now, we plug in the condition for firm profits:

$$C_t + NFA_{t+1} + \frac{p_{t+1}^H}{P_{t+1}} k_{t+1} + b_{t+1} = (1 - \tau^w) w_t N_t + \kappa_t^{ST,H} U_t^{ST,H} + \kappa_t^{ST,L} U_t^{ST,L} + \kappa_t^{LT} U_t^{LT} \quad (C.6)$$

$$- \tau_t + (1 + r - \delta)(NFA_t + \frac{p_t^H}{P_t} k_t + b_t) + \overbrace{\frac{p_t^H}{P_t} z_t F(\tilde{k}_t) N_t - \frac{p_t^H}{P_t} r k_t - w_t N_t - \kappa^v V_t}^{d_t}$$

We can cancel out labor income, but labor tax revenue remains. We use the definition for aggregate output, $Y_t = z_t F(\tilde{k}_t) N_t$ and rearrange:

$$\frac{p_t^H}{P_t} Y_t = C_t + NFA_{t+1} + \frac{p_{t+1}^H}{P_{t+1}} k_{t+1} + b_{t+1} + \tau^w w_t N_t + \tau_t - \kappa_t^{ST,H} U_t^{ST,H} - \kappa_t^{ST,L} U_t^{ST,L} - \kappa_t^{LT} U_t^{LT} \quad (C.7)$$

$$+ \frac{p_t^H}{P_t} r k_t + \kappa^v V_t - (1 + r - \delta)(NFA_t + \frac{p_t^H}{P_t} k_t + b_t)$$

Rewriting gives:

$$\frac{p_t^H}{P_t} Y_t = C_t + NFA_{t+1} - (1 + r - \delta) NFA_t + \frac{p_{t+1}^H}{P_{t+1}} k_{t+1} - \frac{p_t^H}{P_t} (1 + r - \delta) k_t \quad (C.8)$$

$$+ b_{t+1} - (1 + r - \delta) b_t + \tau^w w_t N_t + \tau_t - \kappa_t^{ST,H} U_t^{ST,H} - \kappa_t^{ST,L} U_t^{ST,L} - \kappa_t^{LT} U_t^{LT} + \frac{p_t^H}{P_t} r k_t + \kappa^v V_t$$

Now, $\frac{p_t^H}{P_t} r k_t$ combines with $-\frac{p_t^H}{P_t} (1 + r - \delta) k_t$ to yield $-\frac{p_t^H}{P_t} (1 - \delta) k_t$. We then use the definition of investment, the government budget constraint, and the law of motion for net foreign assets to substitute in net exports:

$$\frac{p_t^H}{P_t} Y_t = C_t + \underbrace{NFA_{t+1} - (1 + r - \delta) NFA_t}_{NX_t} + \underbrace{\frac{p_{t+1}^H}{P_{t+1}} k_{t+1} - \frac{p_t^H}{P_t} (1 - \delta) k_t}_{\frac{p_t^H}{P_t} I_t} - \frac{p_t^H}{P_t} r k_t \quad (C.9)$$

$$+ \underbrace{\tau^w w_t N_t + \tau_t + b_{t+1} - (1 + r - \delta) b_t - \kappa_t^{ST,H} U_t^{ST,H} - \kappa_t^{ST,L} U_t^{ST,L} - \kappa_t^{LT} U_t^{LT}}_{GBC=0} + \frac{p_t^H}{P_t} r k_t + \kappa^v V_t$$

Thus,

$$\frac{p_t^H}{P_t} Y_t = C_t + \frac{p_t^H}{P_t} I_t + NX_t + \kappa^v V_t \quad (C.10)$$

We can now use the definitions for consumption as well as net exports:

$$\frac{p_t^H}{P_t} Y_t = \frac{p_t^H}{P_t} c_{H,t} + \frac{p_t^F}{P_t} c_{F,t} + \frac{p_t^H}{P_t} c_{H,t}^* - \frac{p_t^F}{P_t} c_{F,t} + \frac{p_t^H}{P_t} I_t + \kappa^v V_t \quad (C.11)$$

Note that imports of consumption goods cancel out. This gives:

$$\frac{p_t^H}{P_t} Y_t = \frac{p_t^H}{P_t} c_{H,t} + \frac{p_t^H}{P_t} c_{H,t}^* + \frac{p_t^H}{P_t} I_t + \kappa^u V_t \quad (\text{C.12})$$

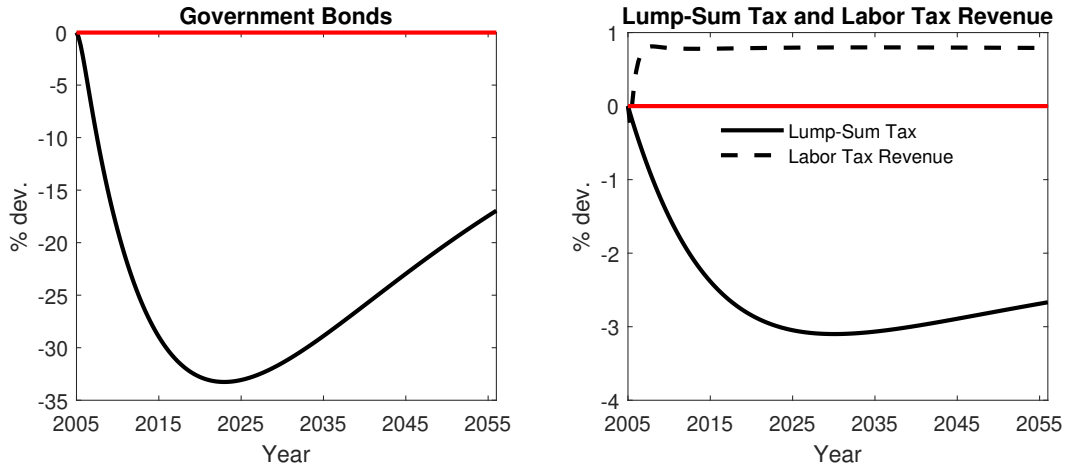
Finally, we rearrange and multiply by the inverse of the price ratio and get

$$Y_t = c_{H,t} + c_{H,t}^* + I_t + \frac{P_t}{p_t^H} \kappa^u V_t. \quad (\text{C.13})$$

C.3 The Government Sector

Figure 17 shows the model responses of the government sector after the reform. The left panel illustrates a decline in government bond issuance and expenditures. The right panel shows higher labor tax revenue and a lower lump-sum tax, consistent with an improved fiscal position driven by higher employment and lower unemployment benefit spending.

Figure 17: Model Responses: The Government Sector



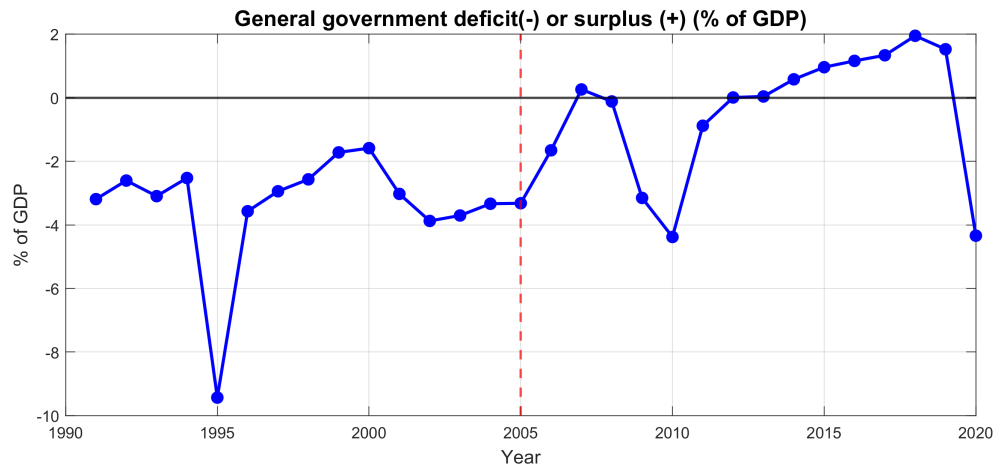
Notes: Model responses in percent deviations from the pre-reform steady state.

The model's predictions are consistent with empirical data. Figure 18 shows the general government deficit or surplus as a percentage of GDP from the early 1990s through 2020. Following the Hartz IV reforms in 2005, as indicated by the vertical line, the deficit clearly decreased, eventually leading to a surplus in subsequent years.

Figure 19 further supports the model's predictions, showing the evolution of German labor tax rates and social security contributions (SSC) for unemployment benefits from 1991 to 2020. The data indicate a significant decrease in the top tax rate and SSC after 2005 (starting in 2007), reducing the financial burden on workers.

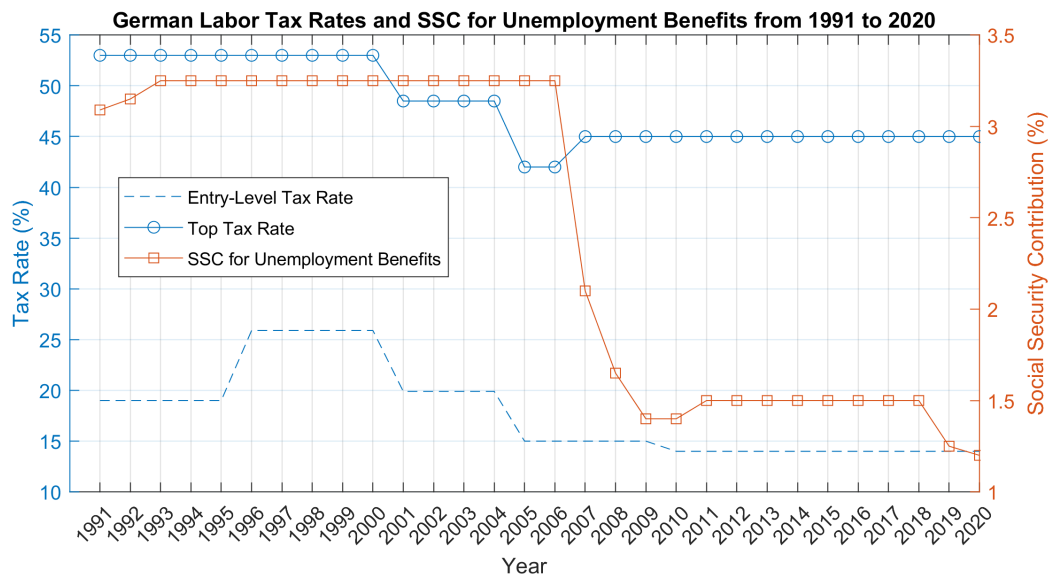
Thus, the empirical evidence on the government deficit, tax rates, and social security contributions aligns with the model's predictions of a decline in government debt, as discussed in Section 5.1.

Figure 18: General Government Deficit(-) or Surplus(+) (% of GDP)



Notes: General government deficit(-) or surplus (+) (ESA 2010) as a percentage of GDP. Data Source: Bundesbank 2024 (BBK01.BK9190).

Figure 19: German Labor Tax Rates and SSC for Unemployment Benefits (1991-2020)



Notes: Evolution of the entry-level tax rate, top tax rate, and social security contributions (SSC) for unemployment benefits in Germany from 1991 to 2020. The left vertical axis shows tax rates as percentages, while the right vertical axis shows the percentage of SSC for unemployment benefits. Data Source: German Federal Ministry of Finance and Federal Ministry of Labour and Social Affairs.

C.4 The Representative Agent Version

The Representative Agent's Problem In the representative agent version of our model, all consumers are fully insured against idiosyncratic risk. The consumption function follows the same CES aggregation as in the baseline:

$$c_t = \left(\gamma^{1-\eta^C} c_{H,t}^{\eta^C} + (1-\gamma)^{1-\eta^C} c_{F,t}^{\eta^C} \right)^{\frac{1}{\eta^C}}.$$

The representative agent's maximization problem is:

$$V_t = \max_{\{c_t, a_{t+1}\}} u(c_t) + \beta \mathbb{E}_t[V_{t+1}].$$

subject to the budget constraint:

$$\begin{aligned} c_t + a_{t+1} = & (1 - \tau_t^w) w_t^L N_t^L + (1 - \tau_t^w) w_t^H N_t^H + \kappa_t^{ST,L} U_t^{ST,L} + \kappa_t^{ST,H} U_t^{ST,H} + \kappa_t^{LT} U_t^{LT} - \tau_t \\ & + (1 + r - \delta) e^{-\psi(NFA_{t-1} - \overline{NFA})} a_t, \end{aligned}$$

where c_t is aggregate consumption, and a_{t+1} is asset holdings for the next period. w_t^L and w_t^H represent wages for low- and high-skilled workers, with N_t^L and N_t^H denoting the respective numbers of employed workers. Short-term unemployed workers in skill groups L and H receive benefits $\kappa_t^{ST,L}$ and $\kappa_t^{ST,H}$, while long-term unemployed workers receive κ_t^{LT} . Taxes are levied at a lump-sum rate τ_t , and net foreign assets evolve according to a debt-elastic interest rate adjustment, with NFA_t denoting the country's foreign asset position. Note that to ensure stationarity of net foreign assets, we follow [Schmitt-Grohe and Uribe \(2003\)](#) and assume a debt-elastic interest rate on international bonds, where the effective return depends on deviations of the net foreign asset position from $\overline{NFA}$. This ensures stationarity of net foreign assets and is not required in our benchmark model. We set $\psi = 0.00074$ for the simulation, following [Schmitt-Grohe and Uribe \(2003\)](#).⁵⁷

The first-order conditions for the representative agent's optimization problem are:

$$\begin{aligned} c_t : \lambda_t &= c_t^{-\sigma} \\ a_{t+1} : \lambda_t &= \beta \lambda_{t+1} (1 + r - \delta) e^{-\psi(NFA_t - \overline{NFA})} \end{aligned}$$

Rearranging the condition for asset holdings gives the Euler equation for optimal intertemporal consumption:

$$\lambda_t = \beta \lambda_{t+1} \cdot (1 + r - \delta) \cdot e^{-\psi(NFA_t - \overline{NFA})}$$

⁵⁷ This prevents long-run effects on the NFA position by construction.

where λ_t represents the marginal utility of consumption, $\overline{NFA}$ is the steady state level of net foreign assets, and ψ is the adjustment cost parameter.

Wage Setting In the heterogeneous agent model, the utility function includes the average consumption of each worker type. In contrast, the representative agent model assumes perfect consumption pooling, leading to identical consumption levels across all workers. Consequently, wages and unemployment benefits directly enter the utility function, replacing worker-specific consumption levels in the bargaining process.

The value of working for a low-skilled consumer is

$$\mathcal{W}_t^{E,L} = u(w_t^{E,L}) + \beta \mathbb{E}_t [(1-s^L)(1-\pi^E)\mathcal{W}_{t+1}^{E,L} + (1-s^L)\pi^E\mathcal{W}_{t+1}^{E,H} + s^L\mathcal{W}_{t+1}^{ST,L}], \quad (\text{C.14})$$

and for a high-skilled consumer, the utility of working is given by

$$\mathcal{W}_t^{E,H} = u(w_t^{E,H}) + \beta \mathbb{E}_t [(1-s^H)\mathcal{W}_{t+1}^{E,H} + s^H\mathcal{W}_{t+1}^{ST,H}]. \quad (\text{C.15})$$

For short-term unemployed consumers with skill level $j \in (L, H)$, the value of being short-term unemployed is

$$\mathcal{W}_t^{ST,j} = u(\kappa_t^{ST,j}) + \beta \mathbb{E}_t [\lambda_t^w \mathcal{W}_{t+1}^{E,L} + (1-\lambda_t^w)(1-\pi^{LT})\mathcal{W}_{t+1}^{ST,j} + (1-\lambda_t^w)\pi^{LT}\mathcal{W}_{t+1}^{LT}]. \quad (\text{C.16})$$

And the value of being long-term unemployed is given by

$$\mathcal{W}_t^{LT} = u(\kappa_t^{LT}) + \beta \mathbb{E}_t [\lambda_t^w \mathcal{W}_{t+1}^{E,L} + (1-\lambda_t^w)\mathcal{W}_{t+1}^{LT}]. \quad (\text{C.17})$$

Note that the union's surplus and the wage-sharing rule remain identical to our benchmark incomplete markets model.

Net Foreign Asset Dynamics The dynamics of net foreign assets are governed by the following equation:

$$NFA_t = (1+r-\delta) \cdot e^{-\psi(NFA_{t-1}-\overline{NFA})} NFA_{t-1} + NX_t$$

where NX_t represents net exports, and the debt-elastic interest rate mechanism ensures stationarity of the net foreign asset position.

The rest of the representative agent version is identical to the heterogeneous agent model.

D Further Model Results

D.1 Regression Results from Empirical and Simulated Data

Table 11 reports regression estimates from the household-level data and from simulated panel data generated by the model. In the calibration, we choose the decline in the long-term unemployed replacement rate so that the simulated regression reproduces the empirical post-reform savings-rate effect captured by the Hartz IV-Dummy (our targeted moment). Beyond the targeted coefficient, most controls display the same qualitative relationship with household savings in both datasets, while differences in magnitudes reflect unit scaling in the simulated economy and the fact that some aggregate covariates are model-specific. The empirical regression additionally includes household controls X_{it} and the full set of macro controls Z_t as described in Section 2.

Table 11: Regression Results from Empirical and Simulated Data: Aggregate Reform Effects

	GSOEP Savings Rate	Model Savings Rate
Hartz-Dummy	0.010*** (0.002)	0.010*** (0.001)
Time Trend	-0.002*** (0.000)	-0.000** (0.000)
Unemployment rate	-0.003*** (0.001)	-5.187*** (1.490)
Debt/GDP ratio	-0.001*** (0.000)	-0.130*** (0.045)
Terms of Trade	-0.002*** (0.000)	0.845** (0.373)
GDP growth	0.000 (0.000)	
GDP		-3.004*** (0.831)
Constant	0.406*** (0.029)	8.920*** (2.430)
Observations	125701	4000000
HH Fixed Effects	Yes	Yes

Notes: All regressions include individual fixed effects. The empirical model corresponds to specification 2.1 in section 2. “Household controls X_{it} and the full macro control vector Z_t are included in the empirical specification but not reported. To retain the first period, the model-based regression uses GDP in levels rather than GDP growth. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.2 Steady-state Comparison

Table 12 displays the percentage changes between the initial (pre-reform) steady state and the steady state following the complete reform package. Wages for low-skilled workers decrease slightly, while wages for high-skilled workers increase slightly, driven by rising market tightness. Additionally, the prices of domestically produced goods and the consumer price index (CPI) experience a slight increase, reflecting higher long-run domestic demand.

The reform positively impacts the labor market by significantly increasing market tightness and the job-finding rate while reducing unemployment in the post-reform steady state. The number of short-term unemployment benefit recipients declines considerably. In contrast, the number of long-term recipients increases due to faster transitions from short-term to long-term benefits following the reduced entitlement duration.

Regarding aggregate variables, we observe higher consumption levels, aggregate assets, and output in the post-reform steady state. The reduction in unemployment benefits triggers consumers' precautionary savings motive, increasing asset holdings. This results in a higher aggregate capital stock and increased investment in the long run. Higher imports accompany the rise in domestic consumption, while exports are slightly lower in the post-reform steady state. Consistent with higher desired saving and bond-market clearing, the net foreign asset position (as a share of output) rises by about 40 percentage points.

Table 12: Long-run Reform Effects: Percent Deviation from Pre-reform Steady State

Prices		Percent Deviations from the Pre-reform Steady State
	Wage, Low-skilled	-0.03
	Wage, High-skilled	0.06
	Price of home good	0.15
	CPI	0.10
	Real Exchange Rate	-0.15
Labor Market		
	Market Tightness	13.75
	Vacancies	7.46
	Unemployment	-5.53
	Job-finding rate	6.65
	Job-filling rate	-6.24
	Low-skilled Short-term unemployed	-11.48
	High-skilled Short-term unemployed	-11.48
	Long-term unemployed	8.49
	Employment	0.75
Aggregates		
	Consumption	1.53
	Aggregate Assets	5.43
	Output	0.75
	Investment	0.75
	Exports	-0.39
	Imports	1.82
	Net foreign assets/Y (pp)	39.62

Notes: Long-run reform effects in percent deviations (percentage points if otherwise indicated) from the pre-reform steady state compared to the post-reform steady state in our baseline scenario.

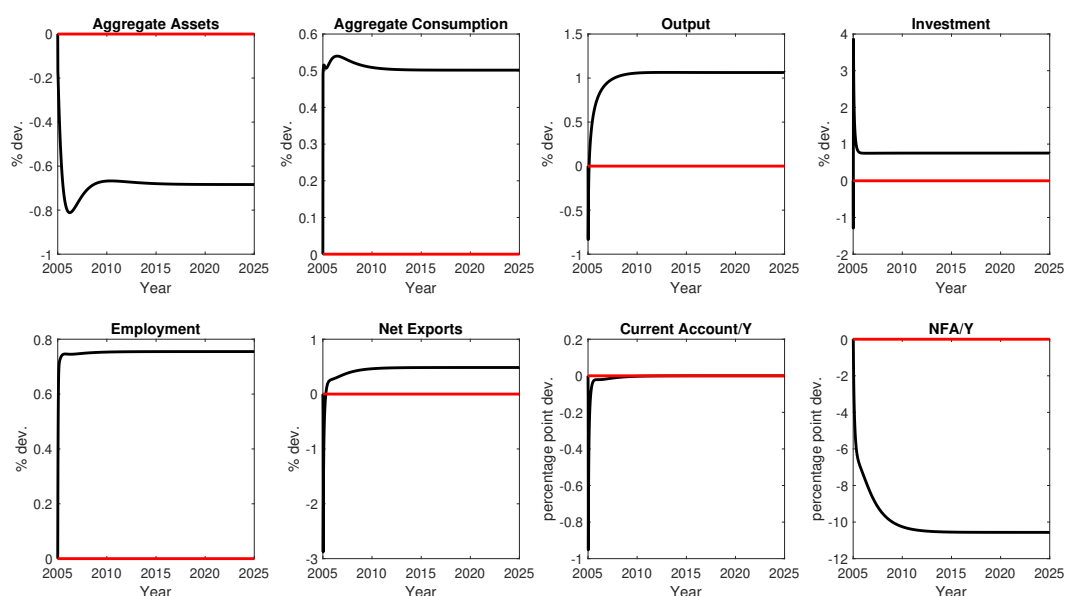
D.3 Increase in Matching Efficiency: The Hartz III Reform

As discussed in Appendix A, the Hartz IV reform was part of a broader agenda to restructure the German labor market. This subsection examines the effects of the Hartz III reform, implemented in 2004, which aimed to restructure the Federal Employment Agency to improve job matching and placement services. We implement the German Hartz III reform by increasing the job-matching efficiency parameter (κ^e) by 7.3 percent, which yields the same steady state decline in unemployment. Figure 20 illustrates the dynamic effects of increasing the matching efficiency along the transition to the new steady state. An increase in matching efficiency leads to positive labor market effects, including higher consumption and output, but they do not trigger a precautionary savings response as consumers' consumption risk in case of unemployment remains

unaffected by a change in the matching efficiency. On the contrary, the expansionary impact of improved job-matching efficiency reduces aggregate assets, leading to a decline in the current account-to-output ratio and net foreign assets. Table 13 presents the corresponding steady state effects of the increased matching efficiency.

It is important to note that our empirical estimation of the effect of Hartz IV on savings presented in Section 2 provides a conservative, lower-bound estimate of the reform's effects as any simultaneous expansionary impacts of other reforms, like Hartz III, would likely have dampened the overall increase in savings. Because Hartz III improved labor market conditions and reduced unemployment, households faced less unemployment risk, reducing their need for precautionary savings. Consequently, Hartz III shifts external balances in the opposite direction by lowering precautionary saving, as improved labor market prospects reduce unemployment risk without changing consumption insurance.

Figure 20: Simulation Results of an Increase in Matching Efficiency (Hartz III Reform)



Notes: Model responses of aggregate variables in percent deviations (percentage point deviations where indicated) from the pre-reform steady state to a 7.3 percent increase in matching efficiency.

D.4 Small open vs. Closed Economy

Figure 21 compares the impulse responses to a labor market reform between a small open economy (solid line) and a closed economy (dashed line). While the labor market impacts are broadly similar across both settings, the closed economy experiences a reduced increase in aggregate assets due to the declining interest rate and the restriction of asset supply to domestic assets. This dampened consumption-savings trade-off results in a smaller initial decline in consumption and a quicker recovery, though it stabilizes at a slightly lower steady state level than in the small open economy. However, the lower interest rate in the closed economy stimulates capital formation, leading to a more pronounced increase in investment and a higher rise in output than in the small open economy.

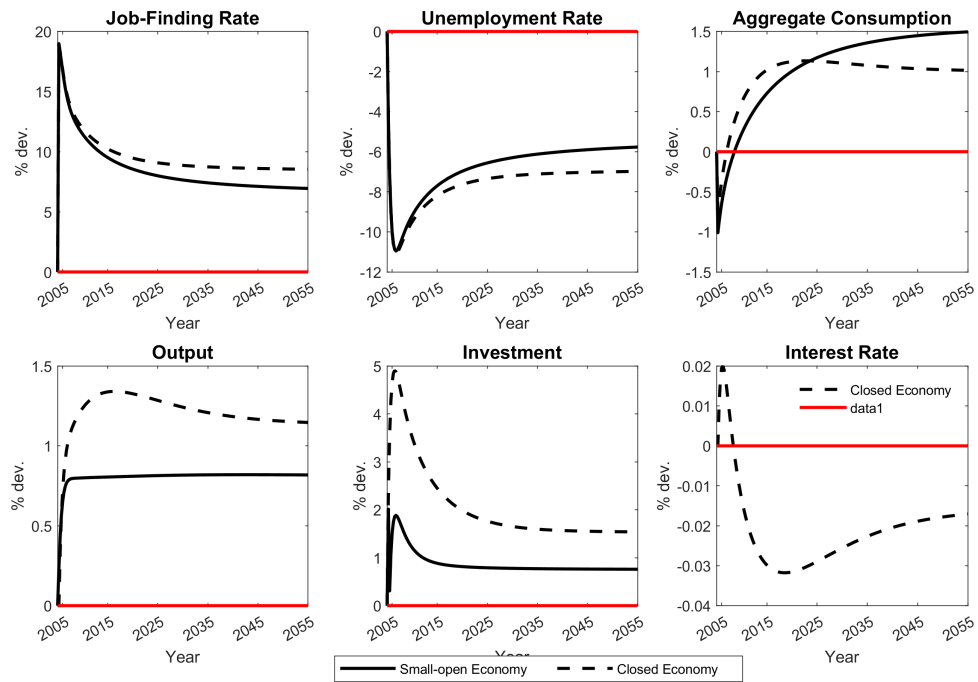
Table 13: Long-run Reform Effects of Hartz IV and Hartz III (Increase in Matching Efficiency): Percent Deviations from the Pre-reform Steady State

	Hartz IV Reform	Hartz III Reform
Prices		
Wage, Low-skilled	-0.03	-0.01
Wage, High-skilled	0.06	-0.03
Price of home good	0.15	-0.24
CPI	0.10	-0.16
Real Exchange Rate	-0.15	0.24
Labor Market		
Market Tightness	13.75	-1.19
Vacancies	7.46	-6.66
Unemployment	-5.53	-5.54
Job-finding rate	6.65	6.66
Job-filling rate	-6.24	7.94
Low-skilled Short-term unemployed	-11.48	-3.21
High-skilled Short-term unemployed	-11.48	-3.21
Long-term unemployed	8.49	-11.03
Employment	0.75	0.75
Aggregates		
Consumption	1.53	0.50
Aggregate Assets	5.43	-0.68
Output	0.75	0.75
Investment	0.75	0.75
Exports	-0.39	0.63
Imports	1.82	0.04
Net foreign assets/Y (pp)	39.62	-10.56

Notes: Hartz IV reform refers to our baseline reform. The Hartz III reform corresponds to an increase in matching efficiency by 7.3 percent (without changes in the replacement rate or entitlement duration); effects are in % dev. from steady state.

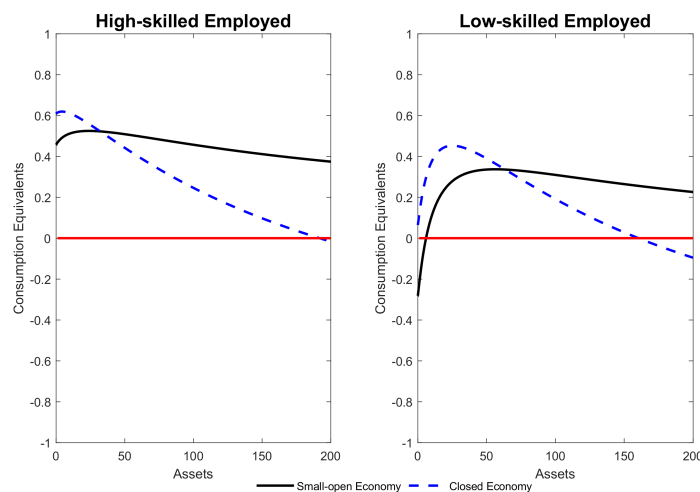
Figures 22 and 23 compare the welfare effects of the labor market reform between a small open economy (solid black line) and a closed economy (dashed blue line) for high-skilled and low-skilled employed workers, measured in consumption equivalents (CE) along the wealth distribution. In a closed economy, the positive labor market effect is more substantial, benefiting asset-poor consumers more than in a small open economy. Additionally, the new steady state features a lower endogenous interest rate, reducing returns on savings. As a result, welfare declines for consumers with very high asset holdings, although the mass of agents at these wealth levels is negligible. Given the high mass of agents with low asset holdings, overall welfare effects are more positive in the closed economy.

Figure 21: Comparison to the Closed Economy Framework



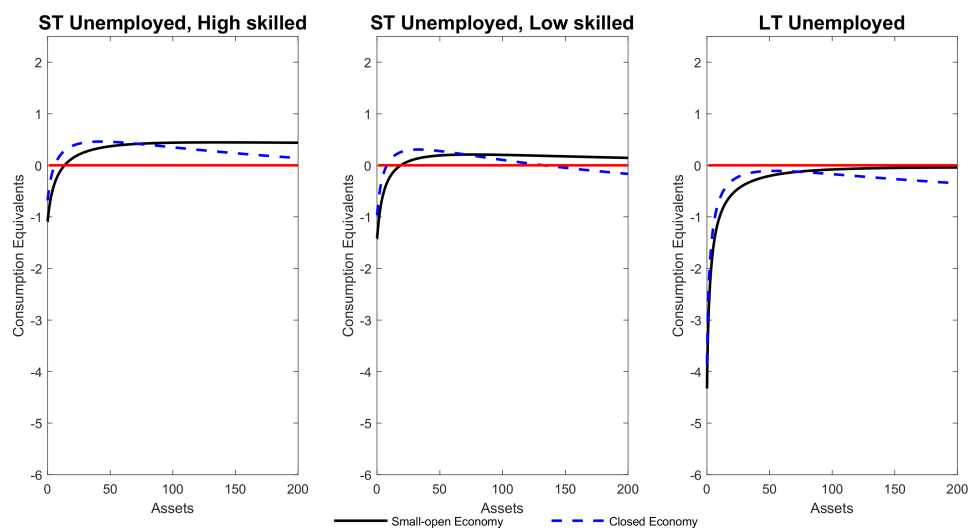
Notes: Responses to the labor market reform in the small open economy (solid line) and the closed economy (dashed line).

Figure 22: Welfare Effects in the Small Open vs. Closed Economy. Employed Workers (Steady state Comparison)



Notes: Consumption Equivalents (CE) of the welfare effects of the full labor market reform in the small open economy (solid black line) and the closed economy (dashed blue line) for employed workers by skill group.

Figure 23: Welfare Effects in the Small Open vs. Closed Economy. Unemployed Workers (Steady state Comparison)



Notes: Consumption Equivalents (CE) of the welfare effects of the full labor market reform in the small open economy (solid black line) and the closed economy (dashed blue line) for unemployed workers by skill group.